



A review on turbulent flow over rough surfaces: Fundamentals and theories

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ABSTRACT

Surface roughness can significantly influence the fluid dynamics and heat transfer in convective flows by inducing perturbations in the velocity profile which affect surface drag, turbulent mixing and heat transfer. While surface roughness can often negatively affect the performance of systems, it can also lead to performance improvements, such as in convective flows where roughness elements have been shown to enhance heat transfer. Turbulent flows over rough surfaces have been studied for about a century leading to significant developments in this field. Direct Numerical Simulation (DNS) has made significant contributions to the knowledge of turbulent flows over rough surfaces as well as evaluation of the developed theories. Moreover, the turbulent closures model has seen wide use for simulation of rough-wall turbulent flows in practical applications where DNS is hindered by its complexity and computational resources. Despite a significant number of experimental and CFD studies and the latest advances in this field, a recent review was not available. Therefore, this review surveys the past and recent experimental and numerical studies to address the fundamentals and theories related to the structure of turbulent flows over rough walls. This study chiefly investigates the structure of mean flow profile over rough surfaces, and its correlation with smooth wall mean flow profile. This review study can contribute to prospective experimental and CFD work, and for characterising rough-wall turbulent flows and heat transfer in different academic and engineering applications such as aerodynamics, hydraulics, meteorology, and manufacturing. The review concludes that despite significant progress, the structure of turbulent flow is still not fully understood. This is mainly due to a lack of systematic studies on the structure of turbulent flow and also due to the variety of roughness which influence the dynamics of the flow in the roughness sublayers. The current roughness scale (sand-grain roughness height) fails to completely characterise roughness in many cases. Therefore, there is a need for a universal roughness scale that can describe every type of roughness and be used in any rough-flow regimes, including fully rough and transitionally rough regimes.

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1. Introduction

The study of roughness in turbulent flows was pioneered by Hagen [1] and Darcy [2] who found that roughness affected pressure drop by increasing the drag force and blockage effects [3]. The formation of the boundary layer over surfaces significantly influences

transport phenomena such as heat and momentum transfer. The roughness of these surfaces can significantly affect the boundary layer flow and thus influence heat and momentum transfer over the fluid-surface interface. Fig. 1 illustrates the boundary layer profile over smooth and rough surfaces. At the leading edge of the plate, flow is laminar; until at some point flow instability occurs initiating a laminar-turbulent transition.

For a turbulent boundary layer over a smooth surface, a thin layer of laminar flow known as the viscous sublayer forms along the entire length of the surface, as shown in Fig. 1 (a). For the same flow over a rough surface Fig. 1 (b), the roughness increases flow instability close to the wall which can lead to increased localised turbulence disrupting the viscous sublayer, thereby generating the roughness sublayer and affecting pressure drop and heat transfer.

Abbreviations: ABL, Atmospheric boundary layer; ACF, Autocorrelation Function; AEH, Attached eddy hypothesis; AM, Additive manufacturing; CFD, Computational fluid dynamics; DNN, Deep neural network; DNS, Direct numerical simulation; ES, Effective slope; GPR, Gaussian process regression; IBL, Internal boundary layer; IEL, Internal equilibrium layer; LES, Large eddy simulation; M-O, Monin-Obukhov; PDF, Probability density function; RANS, Reynolds-Averaged Navier-Stokes; RTS, Rough-to-smooth; STR, Smooth-to-rough; TS, Tollmien-Schlichting

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Nomenclature

A_f	Frontal area of a single roughness element
A_s	Windward wetted surface area of the single roughness element
C	Smooth wall intercept
C_f	Skin friction coefficient
C_s	Roughness constant
c_p	Specific heat capacity
D_t	Tube diameter
D_h	Hydraulic diameter
D_{cf}	Constricted flow diameter
d	Zero-plane displacement
f_{Darcy}	Darcy friction factor
$f_{Darcy,cf}$	Darcy friction factor based on the constricted diameter
$\tilde{F}(s_k), \tilde{G}(ES), G(ES), F(s_k)$	Auxiliary functions
g	Gravitational acceleration
h	Height
k	Roughness height
k_s	Equivalent sand-grain roughness
k_s^+	Roughness Reynolds number, $k_s^+ = k_s U^* / \nu$
k_{Smooth}^+	Lower band of transitionally rough regime
k_{Rough}^+	Upper band of transitionally rough regime
k^+	Roughness Reynolds number based on the mean roughness height
k_{rms}^+	Roughness Reynolds number based on the root mean square of the roughness
k_u	Kurtosis or roughness
$\mathcal{L}$	Prandtl's mixing length
l	Sampling length
L	Monin–Obukhov (Obukhov) length scale
m_θ	θ th moments of the PDF
Nu	Nusselt number
p	Probability Distribution Function (PDF)
Pr	Prandtl number, $Pr = c_p \mu / \gamma$
Pr_t	Turbulent Prandtl number, $Pr_t = \mu_t c_p / \gamma_t$
Re	Reynolds number
Re_θ	Friction Reynolds number
Re_{cf}	Constricted flow Reynolds number
Re_{ind}	Indifference Reynolds number
Re_{crit}	Critical Reynolds number
R_a	Arithmetic mean deviation
R_q	Root-mean-square roughness
R_v	Maximum valley depth below the mean line, within a single sampling length
R_p	Maximum peak height above the mean line, within a single sampling length
R_z	Maximum peak to valley height of the profile
$R_{z(DIN)}$	Ten-point height based on the definition in DIN standard, within a five sampling length

$R_{z(ISO)}$	Ten-point height based on the definition in ISO standard, within a five sampling length
S	Reference surface area before the addition of roughness
S_f	Total frontal area of the roughness
s_k	Skewness of roughness
s_p	Average roughness slope
St	Stanton number, $St = Nu / (Re Pr)$
T	Mean temperature profile
T_∞	Free stream temperature
T_w	Wall temperature
T^+	Dimensionless temperature profile, $T^+ = (T_w - T) / T^*$
ΔT_0^+	Dimensionless terms that accounts for the conductive heat transfer in semi-stagnant fluid in roughness cavities
$\mathbf{V}$	Velocity field
U	Mean velocity
U_∞	Free stream velocity
U_e	Defect velocity
U_b	Bulk flow velocity
U^+	Dimensionless mean velocity, $U^+ = U / U^*$
U_e^+	Dimensionless defect velocity, $U_e^+ = U_e / U^*$
U^*	Friction velocity, $U^* = \sqrt{\tau_w / \rho}$
ΔU^+	Roughness function
u	Streamwise velocity component
v	Normal-wall velocity component
w	Spanwise velocity component
u'	Streamwise component of the fluctuating velocity
v'	Normal-wall component of the fluctuating velocity
w'	Spanwise component of the fluctuating velocity
$u'^2 +, v'^2 +, w'^2 +, u'v' +$	Dimensionless components of Reynolds stress
W	Width
x	Streamwise direction
y	Normal-wall direction
y^+	Dimensionless wall normal-wall direction, $y^+ = y U^* / \nu$
y^*	Height of the internal sublayer
y_0	Hydraulic (or aerodynamic) roughness length
Y	Modified origin, $Y = y - h_0$
y_{CSL}^+	Dimensionless thickness of a conduction sublayer
z	Spanwise direction
Greek letters	
α	Roughness slope angle
α_{rms}	Root-mean-square of local slope
α_f	Mean forward-facing roughness slope angle
β	Correlation length
Δ	Parameter that accounts for the distribution of the roughness element size
δ	Thickness of the boundary layer
δ_e	Thickness of the internal equilibrium layer
δ_i	Thickness of the internal boundary layer
ε	Turbulent dissipation rate

θ	Spectral area of rough surface
κ	von Kármán constant ($\kappa \approx 0.41$).
Λ	Combined solidity and shape parameter of roughness
λ	Solidity parameter of roughness
μ	Dynamic viscosity
ν	Kinematic viscosity
ζ	Monin–Obukhov dimensionless parameter $\zeta = y/L$
Π	A function represents the free turbulence effect on the mean velocity
ρ	Density
τ	Shear stress
τ_w	Wall shear stress
τ_s	Smooth wall shear stress
τ_R	Rough wall shear stress
ϕ_m	Momentum universal functions
ϕ_H	Heat exchange universal functions
χ	Bandwidth parameter, $\chi = \theta m_4/m_2^2$
ψ	Average roughness element spacing
ω	Wake function

Surface roughness may be a result of the manufacturing processes [4] and the normal use of a product (wear and tear). It may also result from mechanisms – including erosion, corrosion and deposition processes such as in gas turbines [5,6], ice accretion on aircraft wings [7–9], deposit in combustion chambers [10], and biofouling on ship hulls and marine machinery [11,12]. These types of roughness can compromise the safety, performance, and efficiency of systems, mainly due to the change in the turbulent structure as well as momentum and energy transfer. For example, in a review study on gas turbines, Bons [5], reported that natural degradation and deposition processes affected the surface roughness enough to reduce the performance of both compressor and turbine.

“Atmospheric Boundary Layer” (ABL) flows is another area, where flow over terrain, plant and urban canopies (Fig. 2 (a)) act as roughness which influences meteorological and local climate predictions [16–20]. Relevant reviews in ABL include Raupach et al. [21] on theoretical and experimental studies of turbulent flow boundary layers, Raupach & Tom [22] and Finnigan [17] on flow studies over plant canopies, and Roth [23] and Barlow and Coceal [24] on ABL over cities. Although they are occasionally compared with the high Reynolds

number limit of industrial rough-wall turbulent flows, the study of geophysical flow is beyond the scope of this review paper.

Intentionally manufactured roughness features are employed for drag reduction and aerodynamic performance improvement [25–30] as well as heat transfer enhancement [30–33]. Dimples on golf balls [34,35], sharkskin denticles on aircraft surfaces [25,36] provide drag reduction, while heat transfer enhancement in roughened solar air heaters was reported by Sharma and Kalamkar [37], Gawande et al. [38], and Alam and Kim [39]. Roughness can improve the wettability of the surface hydrophobicity [40,41] as well as repellency of the hydrophilic surfaces [42]; which in turn, significantly influence the performance of many fluid flow devices such as heat pipes [43,44], desalination system [45–47], etc.

Surface roughness can also be incorporated with other heat transfer enhancement methods such as modified fluids, especially those containing nano-particles. These “nanofluids” have been found to further improve the performance of the heat exchangers [48,49] and as a new generation of heat transfer fluids, have shown great promise in heat transfer enhancement [50,51]. It has been proved that geometrical constraints such as surface roughness can influence the thermal performance of nanofluids including via increased area and eddy effects [51–53].

Recent advances in manufacturing, including Additive Manufacturing (AM), permit the fabrication of engineered rough surfaces as a design parameter, allowing more control of the shape and pattern of the roughness elements that may be difficult to produce by traditional methods [54,55]. The application of AM in the fabrication of thermal management systems and aerospace applications has rapidly increased in recent years and is an active research topic. AM processes can leave irregular and random surface roughness, the measurement of which were reviewed by Townsend [56]. Other researchers [57–60] have studied the effects of AM fabricated rough surfaces on fluid flow and heat transfer.

As well as experimental studies, Computational Fluid Dynamics (CFD) has been widely adopted for rough flow studies. Surface roughness can be accounted for in CFD by either including roughness effects in turbulent CFD models, such as wall-modelled LES or Reynolds-Averaged Navier-Stokes (RANS), or direct resolving via using DNS or LES. The adoption of a suitable numerical method depends on the physics of the problem (Geometry, Reynolds number, etc.) and more importantly, the available computational resources. Modern computing capabilities have enabled researchers to perform Direct Numerical Simulation (DNS) of turbulent flows over rough surfaces with complex geometries, including irregular/random roughness [61–69] or scanned real surfaces [10,70–75].

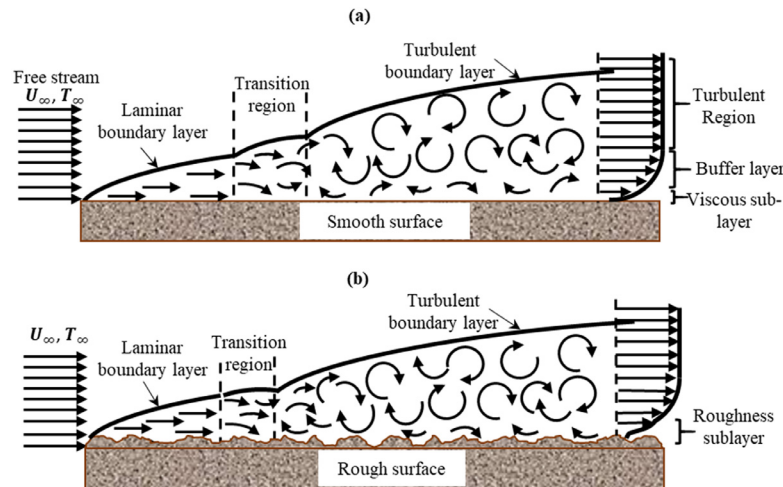


Fig. 1. Boundary layer flows over (a) smooth and (b) rough surfaces.

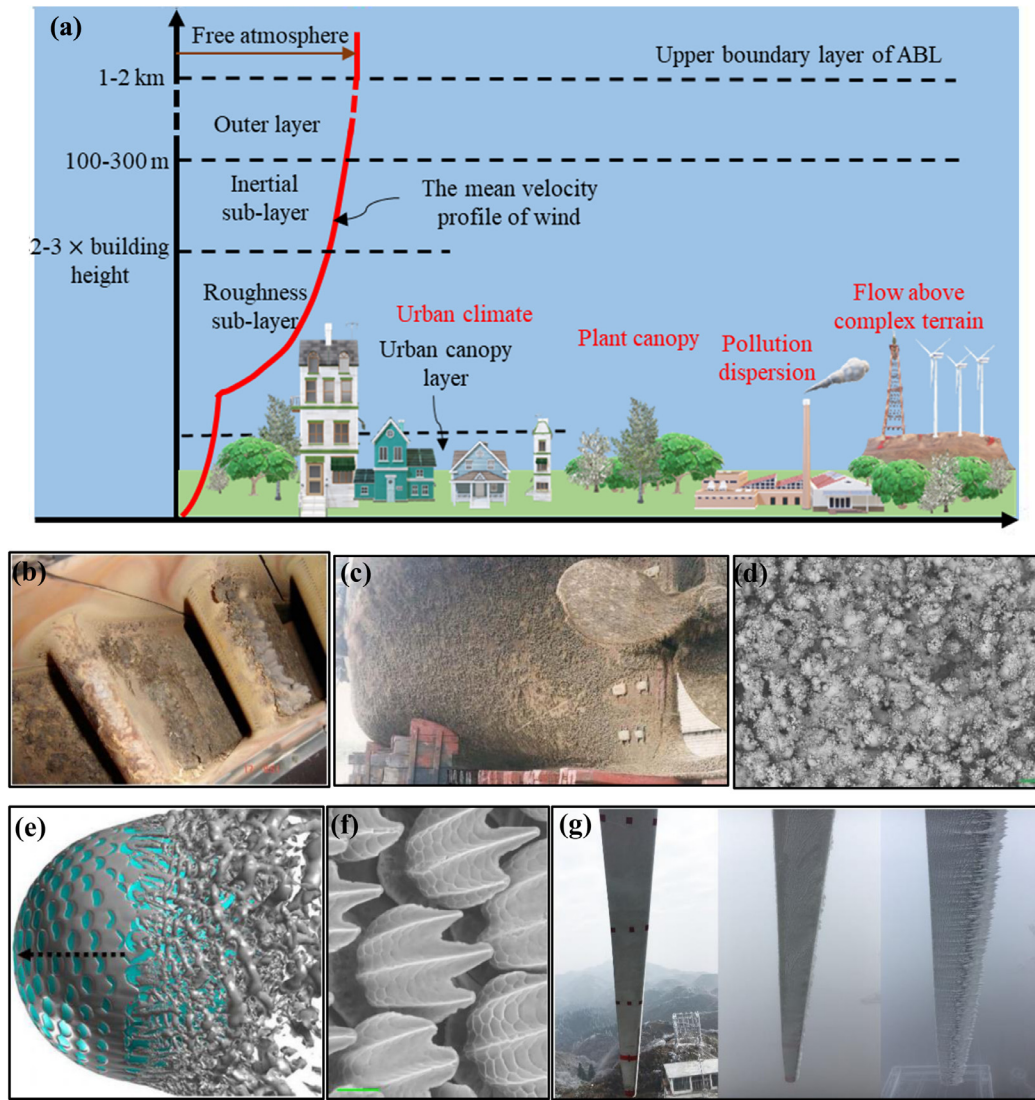


Fig. 2. Various type of roughness; (a) atmospheric boundary layer flows; (b) deposition on gas turbine blades, Bons [5], with permission of Elsevier; (c) biofouling on ship hulls and its impeller, courtesy Yachting Pages; (d) Scanning Electron Microscope image of roughness by laser-based additive manufacturing (green scale bar, $200\mu\text{m}$); (e) Large Eddy Simulation (LES) of flow around a golf ball with dimples by Li et al. [13], with permission of Springer; (f) SEM image of sharkskin denticles (green scale bar, $50\mu\text{m}$) by Oeffner and Lauder [14], with permission of Biologists; (g) ice accretion on wind turbine blades by Shu et al. [15], with permission of Elsevier.

Several textbooks include information on flow and heat transfer over surface roughness such as by Schlichting [76], Wilcox [77] and Kays and Crawford [78]. Despite a large number of experimental and CFD studies on the effect of surface roughness on flow and heat transfer, the number of comprehensive review papers related to this subject is surprisingly low, and most of them were published decades ago. In 1985, Patel et al. [79] reviewed turbulent models for near-wall, and low Reynolds number flows with a small emphasis on surface roughness. Subsequently, in 1991, Raupach et al. [21] reviewed the experimental and theoretical findings of both laboratory and experimental studies on ABL flows. In 1998, Patel [80] reviewed the studies on high Reynolds number flows over rough surfaces for prospective CFD applications. He presented the strength and weakness of some CFD turbulent models for flow over rough surfaces and asserted the need for further experimental studies and more accurate numerical methods for flow over surface roughness. In 2004, Jiménez [81] surveyed the experimental studies related to the characterisation of turbulent flows over different type of roughness. In 2010, Bons reviewed the past 60 years of studies on the effects of surface roughness on gas turbine performance. In the same year, Flack and Schultz [82] surveyed predictive methods used to determine the

frictional drag on rough surfaces. More recently, in 2019, Piomelli [83] gave a review of the studies conducted with DNS and LES at the Turbulence Simulation and Modelling Laboratory at Queen's University, Canada, emphasizing the usefulness of the data that can be achieved by numerical simulation while espousing the need for a more complete research review on this subject.

From the above survey, a wide gap exists for a comprehensive review of recent and past experimental and numerical studies on the fundamentals and theories of rough-wall turbulent flows. Therefore, this paper provides an overview of past and recent studies in this field in the following structure.

This study chiefly focuses on the difference between the mean flow profile over smooth and rough surfaces; thus, sections are structured as follows. First, a brief background of surface roughness effects on fluid dynamics and heat transfer in large- to nano-scale rough-wall flows is given. A brief review on mean turbulent flow over smooth walls and related theories and their extension to rough-wall flows are presented, including the validity of Townsend's wall similarity hypothesis. This is followed by discussions on the characterisation of roughness and rough-wall flows, comprising a review on d-type and k-type roughness. Subsequently, approaches for the

characterisation of rough flow regimes and related correlations are reviewed. The mean flow over rough surfaces as well as the effect of surface roughness on fluid flow and heat transfer are then discussed.

Finally, this review study can contribute to prospective experimental and CFD research, and for characterising rough-wall turbulent flows that are relevant to broad range of academic and engineering applications such as aerodynamics, hydraulics, meteorology, manufacturing etc.

Hereafter, in this study, the velocity field $\mathbf{V} = u, v, w$ has the components of u , v and w for the streamwise (x), normal-wall (y), and spanwise (z) directions respectively with the mean velocity being represented by U . Any quantity with the superscript “+” denotes the dimensionless wall quantities (for example $U^+ = U/U^*$ and $y^+ = yU^*/\nu$), where ν is the fluid viscosity. The friction velocity is also defined as $U^* = \sqrt{\tau_w/\rho}$ with τ_w the wall shear stress and ρ the fluid density. Note that the friction velocity is also referred to as “wall friction velocity”, “shear velocity”, or “wall shear velocity”. The thickness of the boundary layer is represented by δ while k , k_s , $k_s^+ = k_s U^*/\nu$ represent the roughness height, equivalent sand-grain roughness height (see Section 4.1), and roughness Reynolds number (see Section 4.2), respectively.

2. Structure of wall-bounded turbulent flows

2.1. Mean flow over smooth walls

Turbulent flows are irregular in nature, and the variations from one instant to another is substantial; however, the bulk flow can be described by mean flow characteristics. Any expressions and law for the description of the mean flow must, not only express the mean flow pattern but also satisfy the non-slip boundary condition at the wall and the condition of the zero-slope of velocity at the edge of the boundary layer. The bulk properties of mean flow in both smooth- and rough-wall boundary layers can be derived by classical asymptotic matching process [21,84,85]. Let δ be the thickness of boundary layer and u^* the friction velocity ($\sqrt{\tau_w/\rho}$) (with τ_w the wall shear stress and ρ the fluid density). Assume also the flow is in the

“equilibrium” state, in which δ and u^* have a sufficient slow variation in the streamwise direction x . Therefore, both δ and u^* can be considered as local scale at any particular x direction. The asymptotic matching analysis suggests that the boundary layer (over both smooth and rough walls) comprises of two overlapping regions; an outer layer scaling with u^* and δ , and an inner layer scaling with u^* and a set of length scales characterising the surface. For smooth surfaces, the set of length scales includes just the viscous length scale ν/u^* , while, for rough surfaces, it consists of ν/u^* in addition to all length scales required to completely characterise the roughness (e.g. roughness height, kurtosis, skewness, etc.).

The damping effect of the wall alters the regime of the turbulent flow close to the wall and leads to the formation of the boundary layer flow over the wall into several distinct layers. Fig. 3 illustrates the variation of the dimensionless mean velocity profile of the turbulent boundary layer with the logarithmic distance from the wall. The turbulent boundary layer consists of an “inner layer” which is attached to the wall and the “outer layer” that is generated above the inner layer. The inner layer is associated with the active region of turbulent flow that is affected by viscosity and local flow parameters. In contrast, the outer layer is associated with the inactive turbulent region that is influenced by the wall geometry. The inner region itself consists of three subsequence layers: “viscous (laminar) sublayer”, “buffer layer”, and “logarithmic layer”.

In the viscous (laminar) sublayer, which is attached to the surface, the mean velocity can be expressed as [86]

$$U^+ = y^+; \quad (1)$$

The aptly named log-layer region comprises a natural logarithmic velocity distribution, which is referred to as the “law of the wall” (also termed as “log-law”) and expressed as:

$$U^+ = \frac{1}{\kappa} \ln(y^+) + C, \quad (2)$$

where κ is the von Karman constant ($\kappa \approx 0.41$), and the value of constant C is approximately 5.1. The log-layer is a fully turbulent layer in which the Reynolds shear stress greatly exceeds the viscous shear stress. There is a transition layer, known as the “buffer layer” between viscous sublayer and log layer where viscous shear stress almost equals the Reynolds shear stress ($-\rho u'v'$). The buffer layer is characterised by a high-level turbulent mixing due to turbulent production in this region and increased turbulence intensity. The log-layer region is also characterised by turbulence intensity and production, but to a lesser degree, since it has a smaller velocity gradient compared to the buffer layer.

The logarithmic law is not applicable in the outer layer ($y^+ > 700$) since the measured turbulent velocity exceeds the predicted values. The outer layer is unaffected by the direct influence of the wall, which extended to 70% or 80% of the total thickness of the boundary layer. In this region, free shear layer prevails, and the velocity profile deviates considerably from the logarithmic form. The outer layer is also called the defect layer because there is some apparent velocity defect with respect to the free-stream velocity.

In order to cover the defect layer, Coles [90] suggested that the standard velocity profile of a turbulent boundary layer, with the linear and logarithmic sublayer, could be modified with an additional wake function that comes into effect gradually as the outer edge of the boundary layer approaches. The Coles’ “law-of-the-wake” is written as

$$U^+ = \frac{1}{\kappa} \ln(y^+) + C + \frac{\Pi(x)}{\kappa} \omega\left(\frac{y}{\delta}\right). \quad (3)$$

Coles’ “law-of-the-wake” is a superposition of the law-of-the-wall due to the wall shear stress and the law of the wake due to the free turbulence at the core. Wake functions ω identify how wake modification emerges with increasing y . Coles [90] expressed the wake function numerically, and [91] approximated it as a sine-squared

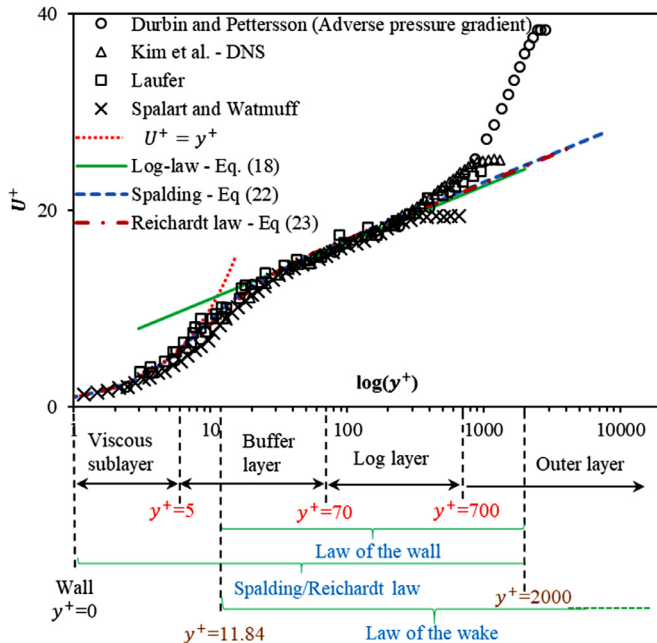


Fig. 3. Mean velocity profile for zero and adverse pressure gradient boundary layer flow. Experimental and DNS data were taken from Durbin and Pettersson [86], Kim et al. [87], Laufer [88], and Spalart and Watmuff [89].

function.

$$\omega\left(\frac{y}{\delta}\right) = \frac{2\Pi}{\kappa} \left(\sin \frac{\pi y}{2\delta}\right)^2. \quad (4)$$

The Π function represents the free turbulence effect on the mean velocity, which may vary along the streamwise direction x . The Π function depends on the pressure gradient and describes how retarded the boundary layer is and how well-developed the wake region is. Usually, the wake function is normalized such that $\omega(1) = 2$, and thus the Π function characterizes the strength of the wake in terms of the deviation from the log law at $y = \delta$ [92,93]. Chauhan et al. [92] suggested a composite profile fit comprising exponential functions that enable calculations of δ and Π for boundary layer. The adapted versions of these profiles were employed by Nagib & Chauhan [94] for pipe and channel flows. An alternative versions of the law-of-the-wake have been proposed by several researchers [95–104].

The simple law-of-the-wall (the log-law) does not satisfy the no-slip boundary condition at the wall and the condition of zero-slope of the velocity profile at the outer edge of the boundary layer. The law-of-the-wake also does not satisfy the no-slip boundary condition. Spalding [105] proposed a single equation that covers all four layers (viscous, buffer, log law, and defect layers) and has reported excellent agreement with experimental data in the range of $0 \leq y^+ \leq 2000$ [106,107]. The Spalding law-of-the-wall is expressed as

$$y^+ = U^+ + \exp(-\kappa c) \left[\exp(\kappa U^+) - 1 - \sum_{n=1}^4 \frac{(\kappa U^+)^n}{n!} \right] \text{ for } 0 \leq y^+ \leq 2000. \quad (5)$$

where the recommended values for κ and c are 0.4 and 5.5, respectively.

Reichardt [108] suggested a correlation for the mean velocity profile of based on the turbulent flow data in pipe and channels, which represent the velocity distribution close to the wall as well as in the core of flow, expressed as

$$U^+ = \frac{1}{\kappa} \ln(1 + \kappa y^+) + 7.8 \left[1 - \exp\left(-\frac{y^+}{11}\right) - \left(\frac{y^+}{11}\right) \exp(-0.33y^+) \right] \quad (6)$$

The Spalding law and Reichardt law can be applied to predict incompressible turbulent flow without adverse pressure gradient and separation. Moreover, both of them satisfy the no-slip condition; however, the zero slope of velocity profile at the outer edge of the boundary layer is not fulfilled, as shown in Fig. 3. Kestin and Richardson [109] summarise the mean velocity profile in the inner region of the turbulent boundary layer.

Turbulent heat transfer over smooth surfaces can be determined by the Reynolds analogy. Based on the similarity between hydraulic and thermal boundary layer, Reynolds [110] postulates the existence of an analogy between wall shear stress, τ_w , and heat flux, q_w . More

discussion on the Reynolds analogy and temperature profile over smooth walls can be found in Bejan [111] and Kays and Crawford [78].

Several hypotheses have been developed to explain the structure of the turbulent boundary layer. Ludwig Prandtl hypothesised the first theory in this field, which is referred to as Prandtl's mixing length theory [113]. The mixing length theory states that the horizontal component of mean flow is proportional to the logarithm of height. Prandtl's mixing length theory assumes that the fluid fluctuations can be simplified and displaced by a distance $\mathcal{L}$, known as Prandtl's mixing length, perpendicular to the main flow direction. Accordingly, the turbulence model of the mixing length can be described by the following equation [76]

$$\tau_t = \rho \ell^2 \left| \frac{\partial U}{\partial y} \right| \frac{\partial U}{\partial y}, \quad (7)$$

where τ_t is turbulent shear stress which is a model of the shear stress created by turbulent fluctuations. Subsequent to Prandtl's mixing length theory, two important hypotheses, the Monin–Obukhov (M–O) similarity hypothesis, and Townsend hypothesis were developed and are now discussed in the following sections.

2.2. Monin–Obukhov (M–O) similarity

The M–O Similarity was originally proposed based on the important understanding of Atmospheric Boundary Layer (ABL) flows. This hypothesis describes the atmospheric boundary layer by Monin–Obukhov (also referred as to Obukhov) length scale, L , expressed as [114–119]

$$L = - \frac{(U^*)^3}{\kappa \left(\frac{g}{T_w} \right) \left(\frac{q}{\rho c_p} \right)}, \quad (8)$$

where κ is the von Karman constant and U^* is friction velocity and g is gravity acceleration, while T_w , q , ρ , and c_p are the wall temperature, kinematic heat flux, fluid density, and fluid heat capacity, respectively. M–O similarity stated that the local variable of turbulent flow could be characterised by “universal functions”. In accordance with the Monin–Obukhov dimensionless parameter $\zeta = y/L$ (where y is the direction normal to the surface), the velocity and temperature profile can be expressed as [120,121]:

$$\left(\frac{\kappa y}{U^*} \right) \frac{\partial U}{\partial y} = \varphi_m(\zeta); \quad (9)$$

$$\left(\frac{y}{T_w} \right) \frac{\partial T}{\partial y} = \varphi_H(\zeta); \quad (10)$$

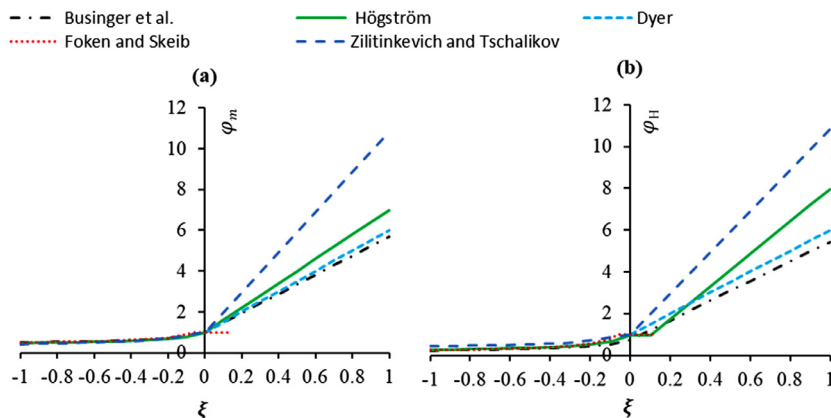


Fig. 4. Universal functions (a) Momentum and (b) heat exchange according to Businger et al. [127], Högström [128] (re-formulated Businger et al. [127]), Dyer [131], Foken and Skeib [132] and Zilitinkevich and Tschalikov [126].

where U and T are mean velocity and mean temperature profiles, respectively, while ϕ_m and ϕ_H are momentum and heat exchange universal functions, respectively. Based on experimental observations, several universal functions have been developed in the literature [122–129]. More discussions on the universal functions can be found in Maronga and Reuder [130], Blaunstein and Kopeika [121], and Foken [120]. Fig. 4 compares some of the universal functions reported in distinctive studies in the literature and demonstrates that universal functions proposed by different authors are in agreement for $\zeta < 0$ (unstable flow stratification condition) but not for $\zeta > 0$ (stable flow stratification conditions).

Nosov et al. [133], Markowski et al. [134], Guo and Zuo [135], and Blaunstein and Kopeika [121] presented some of the recent experiments in this field. Although M-O similarity is universally accepted, despite the validation by field experiments and recent theoretical justifications of the M-O universal functions [136–140], the universality of similarity functions is questionable [141]. In the literature, numerical studies of the validity of the M-O similarity have been performed using LES [130,142–146] and DNS [144,147–153]. Many of these studies calculated universal functions and compared them with predicted ones in the literature, and demonstrated a large scatter for ϕ_m compared to that of ϕ_H .

Foken [129] reviewed the history, experiments, and findings related to M-O similarity. He reported that this theory (even under ideal conditions) has an accuracy of only about 10–20% (typically associated with the universal functions). He also mentioned that M-O similarity is valid mostly within the surface layer, where the turbulent fluxes do not vary more than approximately 10 % of their surface value. McNaughton and Brunet [141] stated that M-O similarity's failure is due to its incomplete basis of parameters since the Monin–Obukhov length scale itself depends on the free stream flow parameters. The M-O similarity is limited to the surface layer and above the roughness sublayer (see Fig. 2 (a)) and probably is not valid for large roughness [129,131]. Recently, Yang et al. [154] reported that the M-O similarity provides good predictions of the mean flow characteristics; however, its evaluation of the turbulent fluctuations is far from satisfactory. Tong and Nguyen [155] showed that the incomplete similarity is related to two assumptions used in M-O similarity: (i) the length scale of the energy-containing eddies only scales with y , and (2) L is only a vertical length scale. Multi-point Monin–Obukhov similarity (MMO similarity) was proposed to address the incomplete similarity in the M-O hypothesis [155–158]. More information on MMO similarity can be found elsewhere [155–159].

M-O similarity has received significant attention for predicting the ABL, especially in most state-of-the-art LES models. It issued as a surface boundary condition to calculate the surface Reynolds stress and the turbulent surface sensible and latent heat fluxes, for example, see Ref. [160–162]. A wide range of studies has employed M-O similarity for considering the effect of the surface roughness in LES (for example, see Ref. [163–167]). Several researchers have pointed out that M-O similarity can only be considered valid within the internal layer (see Fig. 2 (a)) and the upper part of the surface layer; however, it fails in the roughness sublayer (see Fig. 2 (a)), where the direct effect of the roughness elements are presented [168–172]. For an unstable condition, the height of internal sublayer y_* (top of the roughness sublayer) is corresponded to $y_* = 35y_0$, where y_0 is the hydrodynamic roughness length [168]. However, Physick and Garratt [170] suggested $y_* = 50y_0$ and noted that value of y_* might be smaller for stable condition. Garratt [168] and Harman and Finnigan [171] suggested universal functions for below y_* so as to accommodate the roughness sublayer. Based on the previous studies, Basu and Lacser [172] suggested employing $y_* = 50y_0$ as a general rule for LES models, which also used in wall-modelled LES by Maronga et al. [166]. Recently Couvreur et al. [165] analysed the results of ten LES codes and reported a substantial sensitivity of surface fluxes to the specified hydrodynamic roughness length.

2.3. Townsend hypothesis

Despite these shortcomings, the M-O similarity implies that flow can be divided into two separate regions that seem to not interact with each other. This observation, as well as observations indicated by Townsend [222] and Bradshaw [223], led to the development of Townsend hypothesis that has been widely employed to describe turbulent flow over smooth and rough surfaces. Based on Townsend [173] and Bradshaw [174] observations in the laboratory boundary layer, Bradshaw [175] suggested that large-scale motions near smooth walls have no noticeable influence on the relation between shear stress and velocity profile near the wall. Townsend [173] proposed a general hypothesis that the near-wall turbulent flow can be divided into two turbulent regions: “active” and “inactive” regions. The former region is involved in momentum transfer while the latter is not, and these two turbulent regions do not interact. Active turbulence is generated by shear force and scales on the local parameters of the flow and is responsible for producing the Reynolds shear stress. On the other hand, inactive turbulence is produced by the high energy process far from the surface and scales on outer layer parameters. Townsend hypothesis implies that the inertial turbulent scales (also termed the energy-containing motions) dominate the active region and beyond are large relative to the viscous length scale. The inertial scales are independent of Reynolds number since the upper part of the active region is a fixed fraction of the boundary layer thickness, typically 0.15δ .

Townsend [176] described a conceptual model through which the kinematics of boundary layer flow can be statistically explained by the hierarchy of geometrically self-similar attached momentum-carrying motions (or eddies) that are randomly distributed in the flow field and internally dominated. Townsend expressed his Attached Eddy Hypothesis (AEH) as:

“The velocity fields of the main eddies, regarded as persistent, organised flow patterns, extend to the wall and, in a sense, they are attached to the wall.” (Townsend [176], p. 153)

AEH explains that the dominant energy-containing motions in the turbulent boundary layer are due to large eddies attached to the wall. These attached eddies have a population density inversely proportional to their height ranging from a height of $O(y)$ to a height of $O(\delta)$. He proposed using the self-similar function, s_1 , to model the velocity field $\mathbf{V}=(u, v, w)$ for an eddy centred at (x_a, y_a, z_a) and superimposed on a uniform background flow as

$$\mathbf{V} = (u, v, w) = s_1 \left[\frac{(x - x_a)}{y_a} + \frac{(y - y_a)}{y_a} + \frac{(z - z_a)}{y_a} \right]. \quad (11)$$

Several studies have completed the original picture by Townsend [177–180], while plenty of studies have supported the existence of attached eddies in wall-bounded turbulent flows [181–192]. Further discussion on self-similar and attached eddies can be found in McKeon [181]. In a further development on AEH, the Attached Eddy Model (AEM) [193,194] was developed based on Townsend's attached eddy hypothesis [176]. The AEM has proven to provide an effective method for predicting velocity statistics as well as a framework for interpreting the energy-containing flow physics at high Reynolds numbers [194]. The AEM permitted Townsend [176] to make predictions of dimensionless quadrant-decomposed Reynolds shear stress in the active region as

$$u'^2 = B_1 - A_1 \ln \frac{y}{\delta}, \quad (12a)$$

$$v'^2 = B_3 \quad (12b)$$

$$w'^2 = B_2 - A_2 \ln \frac{y}{\delta}, \quad (12c)$$

$$u'v' + = B_4 \quad (12d)$$

where A_1, A_2, B_1, B_2, B_3 , and B_4 are constants. AEH suggests a logarithmic law to characterize streamwise and spanwise turbulence intensity in the inertial region. Several experiments at high Reynolds number [195–203] and recent DNS studies [192,204–207] have reported substantial support for the AEH formulation in Eq. (12a) (a–d). More discussions on the velocity components and turbulent statistics in active and inactive regions can be found elsewhere [208].

A few studies have demonstrated similar logarithmic scaling for higher moments of u' [209] and w' [210,211]. The validity of logarithmic decay of the statistical moments with respect to y has been verified by numerous studies [198,200,204,206,209,212,213]. Among all, Hultmark et al. [214] measured the logarithmic scaling of streamwise turbulence intensity in pipe flows at high Reynolds numbers. Yamamoto & Tsuji [212] performed DNS of high Reynolds number turbulent channel flows and showed logarithmic scaling of streamwise turbulence intensity. The active and inactive region can be deemed uncorrelated only if the nonlinear interaction across these motions are ignored, which have been observed to exist [215–219]. However, such interaction will not significantly influence the Reynolds stress and turbulent statistics [208]. More information on AEM can be found in a comprehensive review by Marusic and Monty [194].

2.4. Wall similarity hypothesis

In order to develop engineering models for the turbulent boundary layer over rough surfaces, knowledge of the structure of these type of flows is required. Current engineering models consider the influence of roughness as small perturbations to the smooth wall boundary layer [80]. Therefore, most of the rough-wall turbulent models have been developed based on the extension of smooth-wall turbulent flow data to the rough-walls. The “wall similarity hypothesis” is an extension to Townsend’s Reynolds number similarity hypothesis [176]. Townsend thoroughly described the general structure of turbulent flows in his book [176], by which he explained his hypothesis and also postulated the Reynolds number similarity expressed as:

“while geometrically similar flows are expected to be dynamically similar if their Reynolds numbers are the same, their structures are also very nearly similar for all Reynolds numbers which are large enough to allow (fully) turbulent flow.” (Townsend [176], p. 53)

Based on Townsend’s Reynolds number similarity, the wall similarity hypothesis for turbulent flow over both rough and smooth surfaces as

“outside the roughness (or viscous) sublayer, the turbulent motion in a boundary layer at high Reynolds number are independent of the wall roughness and the viscosity, except for the role of the wall in setting velocity scale U^ , the height $Y=y-d$ and the boundary layer thickness δ .”*

Here, d is the zero-plane displacement. The wall similarity hypothesis is also referred to in the literature as the Townsend Reynolds similarity for rough-walls. Wall similarity was supported with the scaling law for velocity spectra in several overlapping spectral ranges by Perry and Abell [220] and also by Perry and Chong [193] and Perry et al. [221,222]. This hypothesis was later labelled as “wall similarity” by Raupach et al. [21]. The wall similarity hypothesis is also referred to as the outer-layer similarity in the literature.

According to the wall similarity, for sufficiently high Reynolds numbers outside the viscous sublayer, the turbulent motions are

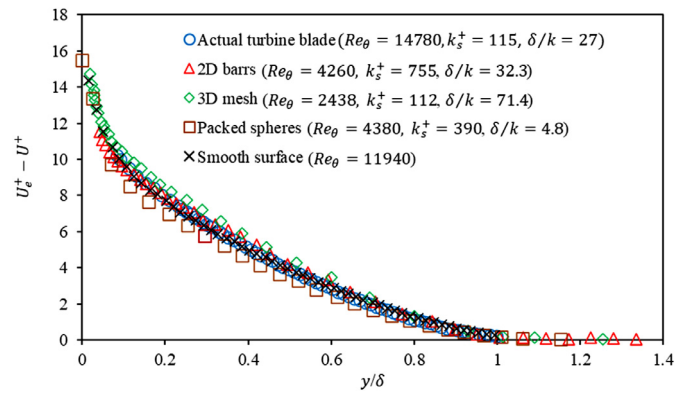


Fig. 5. Mean defect velocity of experimental turbulent flows over 2D bars and 3D mesh by Volino et al. [223], actual turbine blade roughened due to deposition by Wu and Christensen [224], closely packed spheres by Kim et al. [87], and smooth surface by Mejia-Alvarez and Christensen [225].

independent of surface conditions and viscosity (except for wall shear stress and the boundary layer thickness). This means that the turbulent stress, normalised by the wall shear stress, is universal outside the roughness sublayer. The outer region has been generally thought to be unaffected by roughness, as reported in reviews by Raupach et al. [21] and Jiménez [81]. This implies that the mean flow for both smooth and rough walls obeys the universal velocity defect law in the overlap and outer region of the boundary layer, which is expressed as [21]:

$$U_e^+ - U^+ = f\left(\frac{y}{\delta}\right) \quad (13)$$

Fig. 5 illustrates the defect velocity of zero pressure gradient turbulent flow over different roughness. In Fig. 5, Re_θ is the friction Reynolds number that represents the Reynolds number based on the friction velocity. Mean flow similarity can be observed through the overlapping of smooth and rough wall data when plotted in a velocity-defect form.

The wall similarity hypothesis (universal velocity defect law) has been supported by numerous studies with different types of roughness [224,226–240]. The validity of wall similarity hypothesis and defect velocity law has also been verified by numerous experimental studies performed by Karen A. Flack and Michael P. Schultz and their colleagues [82,223,235–237,241–251] and also Kenneth T. Christensen and his colleagues [87,224,225,252–257]. Flack and Schultz [250] outline their 15 years of experimental research on rough-wall turbulent flows, where they reported the observation of wall similarity hypothesis in all their experiments. They also demonstrated that similarity in the outer layer is also observed in the quadrant-decomposed components of Reynolds stress ($u'^2 +, v'^2 +$) and in the turbulent transport of the Reynolds stress ($-u'v' +$). Apart from experimental studies, wall similarity has been validated with DNS of turbulent flow over rough surfaces; including the recent studies of Busse et al. [71], Thakkar et al. [72,73], Busse and Jelly [258], Jayaraman and Khan [259], and Khan and Jayaraman [260].

Some researchers have observed exceptions to wall similarity and outer layer sensitivity to surface condition. These exceptions to wall similarity mostly observed in two-dimensional roughness elements, although recent studies have indicated the validity of wall similarity for two-dimensional roughness [250]. The studies of turbulent flows over woven meshes by Krogstad et al. [261] and transverse bars by Keirsbulck et al. [262] indicated an increase in the wake strength of the mean velocity profile. Krogstad and Antonia [263] reported alterations in the turbulent structure in the outer layer for a woven mesh roughness. Krogstad and Antonia [264] observed variations in the Reynolds shear stress profiles for surface covered with transverse

rods. Schultz et al. [265] indicated that wall effects could have a larger influence on the outer flow with a roughly 40% extension of the attached eddies into the boundary layer, for 2D roughness. Some of the recent DNS studies have also reported a poor match between the smooth and rough wall data in the outer region such as DNS of turbulent flow over streamwise-aligned triangular ridges by Strohh et al. [266]. Lee and Sung [267] performed DNS study of turbulent flows over transverse squared rods [266]. They showed self-preservation of defect velocity can be achieved at about $x > 300\delta_{in}$, where x is the streamwise direction and δ_{in} is the thickness of the boundary layer at the inlet. Chung et al. [268] performed an idealised DNS study to evaluate the wall similarity hypothesis by comparing the effect of no-slip versus shear stress boundary conditions and found that the outer flow is highly insensitive to the imposed wall boundary condition.

Flack et al. [235] outlined the exceptions to wall similarity and explained that these disparate findings are most likely due to “strong” roughness (as they termed) used in the studies. They defined strong roughness as roughness whose equivalent sand-grain roughness height k_s is a significant portion of the inner layer thickness. This argument was also supported in the review by Jiménez [81] which stated that rough-wall flows with $\delta/k \leq 40$ may exhibit roughness effect well into the outer layer due to the destruction of a significant portion of the log-law region by roughness elements. An alternative criterion was reported by Flack et al. [235] as $\delta/k_s < 40$. The woven mesh, traverse rods, and transverse bars used in the studies of Krogstad et al. [261], Krogstad and Antonia [263], Keirbulck et al. [262], and Strohh et al. [266] corresponds to $\delta/k_s \approx 15$, $\delta/k_s \approx 8$, $\delta/k_s \approx 7$, and $\delta/k \approx 6.5$ respectively. Reynolds stresses of flow over distinctive rough surfaces agree well throughout most of the boundary layer and the wall similarity is maintained well outside of $3k_s$ while the differences in higher moment turbulence statistics of different rough-wall boundary layers are confined to $y < 5k_s$, as reported by Flack et al. [235].

Conflicting evidence has also been reported in the literature. In a review study, Flack and Schultz [250] evaluated the similarity hypothesis based on the data of several studies with a range of roughness from where similarity is expected to hold ($\delta/k = 110$) to significantly rougher surfaces ($\delta/k = 16$). They showed excellent agreement between smooth- and rough-wall profiles, indicating similarity in the mean flow. Mean flow similarity for large roughness was also supported by Castro [231] who showed similarity for $\delta/k > 5$ and Wu and Christensen [224] who showed mean flow similarity for ($\delta/k = 27$). Wu and Christensen [252] also observed that the quadrant-decomposed Reynolds shear stress and the Reynolds stress for smooth- and rough-wall cases with $\delta/k > 5$ are in accordance with wall similarity hypothesis.

3. Surface roughness parameters and characterisation

This section briefly discusses surface roughness and the methods for characterisation of roughness from an engineering viewpoint. Generally, the surface texture, also known as surface finish and surface morphology, comprises a series of geometrical irregularities that

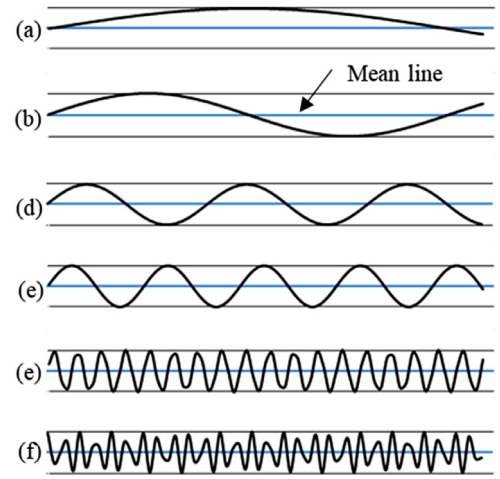


Fig. 6. Surfaces with different wavelengths but the same amplitude. (a) and (b) surface forms; (c) and (d) Waviness; (e) and (f) roughness.

are randomly or uniformly extended over a smooth surface [269]. The frequency of the height of irregularities quantifies the type of surface, as shown in Fig. 6. Geometrical irregularities with a high frequency are considered as “Roughness” while irregularities with a medium frequency are regarded as “Waviness”. Geometrical irregularities with a low frequency are known as “surface form” which is no longer considered as surface texture. The distinction between roughness, waviness and form can be found in Raja et al. [270].

Napoli et al. [61] introduced Effective Slope (ES) to distinguish between roughness and waviness surface, written as

$$ES = \frac{1}{l_x l_z} \int_0^{l_x} \int_0^{l_z} \left| \frac{\partial k(x, z)}{\partial x} \right| dx dz. \quad (14)$$

where $k(x, z)$ is the roughness amplitude, which is also referred to as height map or height function, while l_x and l_z are sampling length in streamwise and spanwise directions, respectively. ES is defined as the average value of the magnitude of the slope of the roughness corrugation. Pyramid roughness and 2D wave-composite surfaces with an effective slope of approximately less than, 0.35, do not scale on the roughness height and are classified as wavy surfaces [61,249]. Yuan and Piomelli [64] found that the critical ES value for real rough surfaces is much larger than 0.35. Forooghi et al. [66] showed that, at lower values of the ES (mild slope), the surface slope becomes the main controlling factor and suppresses the impact of the topographical parameters.

3.1. Roughness parameters

The flow characteristics over rough surfaces should be expressed in terms of statistical surface properties in order to be compatible with the stochastic nature of irregular roughness distribution.

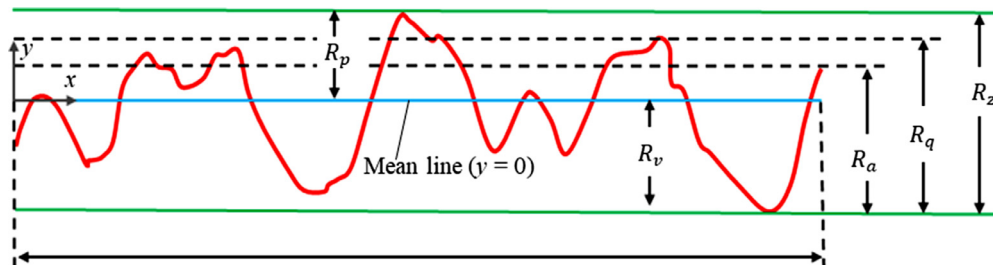


Fig. 7. Schematic representation of R_a and R_q parameters.

Table 1
Definition of roughness parameters.

Parameter	Description	Mathematical description	Discrete implementation
R_a	Arithmetic mean deviation	$\frac{1}{l} \int y dx$	$\frac{1}{n} \sum_{i=1}^n y_i $
R_q	Root-mean-square roughness	$\sqrt{\frac{1}{l} \int y^2 dx}$	$\sqrt{\frac{1}{n} \sum_{i=1}^n y_i^2}$
R_v	Maximum valley depth below the mean line, within a single sampling length	$ \min y(x) $	$ \min y_i $
R_p	Maximum peak height above the mean line, within a single sampling length	$\max y(x)$	$\max y_i$
R_z	Maximum peak to valley height of the profile	$R_v + R_p$	
$R_{z(DIN)}$	Ten-point height based on the definition in DIN standard, within a five sampling length	$\frac{1}{10} (\sum_{j=1}^5 R_{pj} + \sum_{j=1}^5 R_{vj})$	
$R_{z(ISO)}$	Ten-point height based on the definition in ISO standard, within a five sampling length	$\frac{1}{5} (\sum_{j=1}^5 R_{pj} - \sum_{j=1}^5 R_{vj})$	
s_k	skewness	$\frac{1}{(R_q)^3} \frac{1}{l} \int y ^3 dx$	$\frac{1}{n(R_q)^3} \sum_{i=1}^n y_i^3$
k_u	kurtosis	$\frac{1}{(R_q)^4} \frac{1}{l} \int y^4 dx$	$\frac{1}{n(R_q)^4} \sum_{i=1}^n y_i^4$

International standards have adopted numerous statistical surface parameters for the characterisation of surface roughness [271,272]. Examples of these standards are ASME Y14.36M and ISO 1302. A roughness parameter can be calculated either on a profile of a sampling line l (Fig. 7) or on a sampling area A . The parameters defined based on the profile of the sampling length (known as R parameters) are more common while those identified based on the sampling area (known as S parameters) give more significant values.

Amplitude parameters (R or S) can be defined using a heightmap used to characterize the surface based on the vertical deviations of the roughness profile from the mean line. Commonly-used amplitude parameters (based on the profile shown in Fig. 7) are listed in Table 1 [272]. The roughness amplitude y must be centred, meaning that the mean height calculated on the sampling length l is already subtracted from the heights [190] and n is the number of measured points on the surface.

The R_q parameter statistically defines the standard deviation of the heights distribution on the surface and is of more physical significance than R_a . The sole use of the R_a parameter can lead to misinterpretation of the surface characteristics since different surface structures can possess the same R_a value. The R_z parameter is more sensitive to occasional high peaks or deep valleys than that of R_a .

The skewness is a measure of the profile symmetry and describes the shape of the topography height distribution and can be positive, negative or zero. This parameter is sensitive to occasional deep valleys or high peaks. For a surface with symmetrical topography, i.e. a surface with many peaks as valleys such as Gaussian surfaces, the skewness is zero. A surface with valleys filled in or high peaks (such as deposition on surface, porous, and sintered) has positive skewness. In contrast, surfaces with peaks removed or deep valleys (such as

corrosion and scratched) possess a negative skewness. The schematic representation of three types of skewness is illustrated in Fig. 8 (a).

The kurtosis, k_u , is a measure of the sharpness of the surface height distribution. The kurtosis characterises the distribution of the height and is strictly positive. If $k_u < 3$, the distribution curve is referred to as Platykurtic and has relatively few high peaks and low valleys. If $k_u > 3$, the distribution curve is referred to as Leptokurtic and has relatively many high peaks and low valleys. The schematic representation of three types of kurtosis is shown in Fig. 8 (b).

3.2. Characterisation of surface roughness

The characteristics of roughness can be sufficiently described with the utilisation of the Probability Distribution Function (PDF). It represents the probability of a point on the surface that has a height equal to y ; thus it is given as [273]:

$$\int_{-\infty}^{+\infty} p(y) dy = 1 \quad (15)$$

The distribution curves shown in Fig. 8 are examples of the different possible shape of PDF. The shape of the PDF provides useful information about the surface's nature and can be evaluated by its moments. The central ϑ th moment of the PDF with zero mean, m_ϑ , is defined as [273]

$$m_\vartheta = \int_{-\infty}^{+\infty} y^\vartheta p(y) dy \text{ or in discrete form } m_\vartheta = \frac{1}{n} \sum_{i=1}^n y_i^\vartheta \quad (16)$$

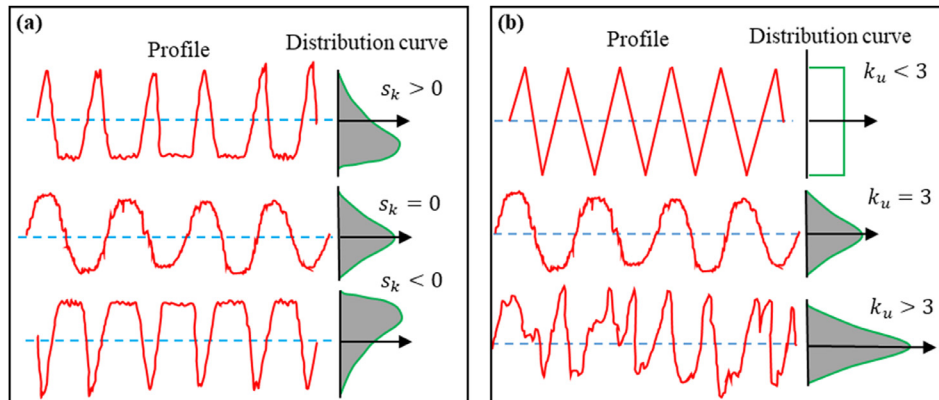


Fig. 8. Different roughness profile and their amplitude distribution curve along with (a) skewness and (b) Kurtosis ranges.

It can be seen that the first moment is zero. The first four moments can adequately describe any statistical distribution of roughness [64,82,224,249,252,274]. The various surface roughness parameters found in the literature can be mathematically determined by using PDF moments. As to the definitions of roughness parameters, described in Table 1, the roughness parameters can be defined as:

$$R_q = \sqrt{m_2}; S_k = m_3/m_2^{3/2}; k_u = m_4/m_2^2 \quad (17)$$

In the past few decades, the characterisation of the roughness scale in turbulent flows has been linked to the statistical moments of the surface PDF up to fourth-order [64,66,275,276]. Recent studies concerning rough-wall flows analysed by the statistical moments up to fourth-order can be found elsewhere [64,82,224,249,252,274].

According to Flack and Schultz [82], a single measure of roughness height, such as R_a or R_q , is not able to fully describe the surface roughness. They showed that the surface roughness could be characterized merely by root-mean-square roughness height and the skewness (or the kurtosis), which have also been supported by other studies [277,278]. Jelly and Busse [279] studied 174 skewness-kurtosis combinations of rough surfaces (natural or synthetic) taken from eleven independent surface measurements published in related literature. They demonstrated that most of the measured data is clustered around the Gaussian case, corresponding to $s_k = 0$ and $k_u = 3$.

Two more important parameters of surface roughness that can significantly influence the rough-wall flows are the solidity parameter λ and the shape parameter. The solidity parameter is defined as the ratio of the average roughness element spacing (ψ) to the effective slope roughness height (k), while the shape parameter is presented as the ratio of the frontal area of a single roughness element (A_f) and the windward wetted surface area of the single roughness element (A_s) [280–282]. A relation of $ES = 2\lambda$ has been observed by several researchers [3,61,72,268]. In a further attempt, Dirling [281] introduced a “combined solidity and shape parameter”, Λ , for 2D roughness, written as

$$\Lambda = \left(\frac{\psi}{k} \right) \left(\frac{A_f}{A_s} \right)^{-4/3} \quad (18)$$

Sigal and Danberg [280] considered the effect of the roughness density for dense arrangements of two-dimensional transverse roughness including bars, rods, and ribs by including the reference surface area before the addition of roughness (S) and the total frontal area of the roughness (S_f), expressed as

$$\Lambda_s = \left(\frac{S}{S_f} \right) \left(\frac{A_f}{A_s} \right)^{-1.6} \quad (19)$$

Since real-life roughness such as turbine blade roughness is random and highly non-uniform, some adoption is needed to apply the definition of Λ_s . For this reason, Bons [283] suggested dividing the surface into several cells and using a 3D surface topography point-by-point to measure the windward and wetted surface areas of each cell of the entire map. Thus, the $\Lambda_{s, \text{modified}}$ can be obtained by modifying the definitions of parameters in Eq. (19) in such a way that S is equivalent to the platform area of the surface map and S_f becomes the summation of the windward projected area of each cell. Moreover, A_f becomes the average windward projected area of all cells, while A_s is the average windward wetted area of all cells.

4. Characterisation of turbulent flow over rough surfaces

As explained in the introduction section, the surface roughness alters the characteristics of turbulent flows. The most significant effect of the wall roughness is increased drag and augmented wall-normal functions near the wall [261,284], which leads to a decrease in the anisotropy of the Reynolds stress [285]. Rough-wall turbulent flows were reviewed by Raupach et al. [21] and Jiménez [81]. They

reported that the boundary layer over both the smooth and rough walls possess quite different structures controlled by different length scales in the inner layer. However, away from the surface, the low-order statistics are reasonably similar, indicating that the outer layer is unaffected by the surface structure (whether it is smooth or rough). Despite long-term research, no universal method for estimation of friction losses has been developed so far; other than hydrodynamic tests or geometrically resolved DNS [286]. However, some empirical correlations for different types of roughness have been developed based on the experimental and DNS studies which are now reviewed in the following sections. [286].

4.1. Equivalent sand-grain roughness

In the derivation of log-law, it is assumed that the near-wall flow is characterised by the viscosity. However, this is merely true for hydrodynamically smooth surfaces where the average height of the surface roughness elements is less than the thickness of the viscous sublayer. Otherwise, the effect of the roughness on the flow should be considered by defining a roughness scale that reflects the effect of roughness characteristics on the flow. In the literature, the equivalent sand-grain roughness height is widely used for characterisation of roughness in rough-wall flows. The equivalent sand-grain approach stems from the pioneer works by Nikuradse [287,288] and Schlichting [3]. The origin of this approach dates from Nikuradse's sand-grain experiments which provided data on the scaling laws. The concept of this approach is based on the definition of a single length scale which can characterise the roughness. This length scale can be applied in various scaling laws for the velocity field and friction factors. Consider that the surface is covered with a layer of uniform packed spheres as dense as possible, as shown in Fig. 9 (a).

The diameter of the sphere is called *sand-grain roughness height* k for the sake of the similarity of the case with sandpaper. For the type of the surfaces physically similar to the sand-grain in Nikuradse's experiment, the length scale can be described by the sand-grain roughness height. However, when the shape of roughness deviates from the sand-grain, a single parameter cannot describe the sand-grain roughness anymore. In this case, geometrical parameters may be applied to characterise these type of surfaces [82]. Therefore, an equivalent length scale was introduced by Colebrook and White [289] referred to as a so-called *equivalent sand-grain roughness* k_s , as shown in Fig. 9 (b). The equivalent sand-grain roughness is an equivalent roughness height for irregular, and non-uniform rough surfaces that produce the same results as the uniform sand-grain of the Nikuradse for a fully rough flow regime. The definition of the fully rough regime is presented in upcoming sections.

4.2. Rough flow regimes

The roughness modifies the law-of-the-wall used for the viscous sublayer. In the context of the law of the wall, the “roughness Reynolds

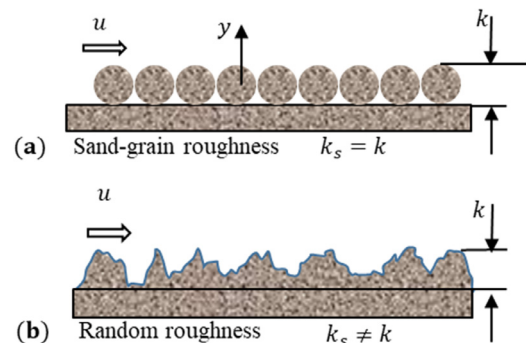


Fig. 9. Equivalent sand-grain roughness.

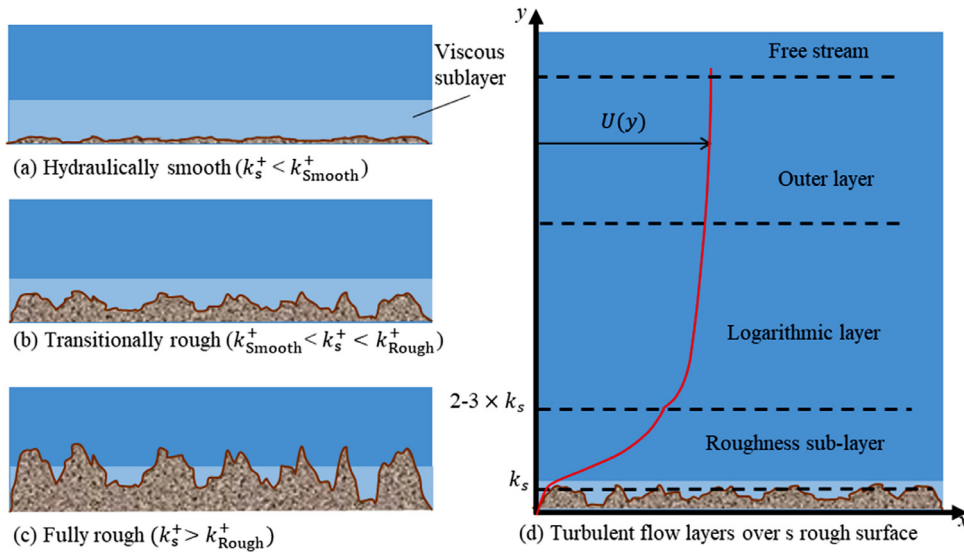


Fig. 10. Turbulent flows over rough surfaces; (a) hydraulically smooth regime with roughness embedded under the viscous sublayer, (b) transitionally rough regime with partially destroyed viscous sublayer, (c) fully rough regime with completely destroyed viscous sublayer, (d) schematic of velocity profile $U(y)$ over roughness with the indication of roughness sublayer, logarithmic layer and outer layer.

number" k_s^+ is defined as

$$k_s^+ = \frac{k_s U^*}{\nu}, \quad (20)$$

Corresponding to the three regions of the wall affected the turbulent boundary layer, the rough flow regimes can be classified into three main categories. Consider k_{Smooth}^+ and k_{Rough}^+ as two critical limits of the roughness Reynolds number. Then, based on the different value for the roughness Reynolds number, three different roughness regimes, are defined as [287,288]

- **Hydraulically smooth regime:** If $k_s^+ \leq k_{Smooth}^+$, the roughness elements are entirely embedded in the viscous sublayer, as shown in Fig. 10 (a); thus, skin friction and drag coefficient are not modified.
- **Transitionally rough regime:** If $k_{Smooth}^+ \leq k_s^+ \leq k_{Rough}^+$, the influence of the surface roughness is complex. In this regime, both the Reynolds number and relative roughness influence the skin friction and drag coefficients. This means that both viscous and pressure forces influence the skin friction and drag coefficient. The *Transitionally rough* regime is associated with the reduced sublayer thickness, as shown in Fig. 10 (b) – resulting in diminishing the influence of the wall damping effect.
- **Fully rough regime:** If $k_s^+ \geq k_{Rough}^+$, the rough elements protrude into the fully turbulent region shifting the logarithmic profile downwards. In this flow regime, as shown in Fig. 10 (c), the viscous sublayer is entirely destroyed by the large turbulence mixing caused by roughness elements. The friction drag significantly increases due to pressure force on the roughness and is independent of the viscous effect (Reynolds number). The pressure loss becomes independent of the molecular viscosity of fluid and velocity when roughness elements intrude viscous sublayer [78].

k_{Smooth}^+ and k_{Rough}^+ are the onsets of transitionally rough and fully rough regimes, respectively. Different values for k_{Smooth}^+ and k_{Rough}^+ , reported in the literature, are listed in Table 2.

The variety of reported values for the onset of the transitionally rough and fully rough regimes for different roughness types obviously indicates the difficulty in definition of scaling parameters applicable for all roughness regimes. Clauser [292] claimed this difficulty on the fact that some roughness types generate roughness functions

Table 2

Values for the lower and upper bands of transitionally rough regime.

Researcher	Roughness type	k_{Smooth}^+	k_{Rough}^+
Nikuradse [287]	Uniform sand-grain	5	70
Ioselevich and Pilipenko [290]	Sand-grain roughness	2.5	90
Ligrani and Moffat [233]	Packed sphere bed	15	50
Langelandsvik et al. [291]	Commercial steel pipes	1.4	18
Shockling et al. [292]	Honed pipe	3.5	30
Schultz and Flack [248]	Surface scratch	2.5	25
Lewkowicz and Musker [293]	Ship-hull roughness (using hk_{ism}^+)	17	40
Cebeci and Bradshaw [294]	Sand-grain roughness	2.25	90

with monotonically changing slope while others possess inflection points.

The entire layer that dynamically influenced by the length scale associated with roughness elements is referred to as “*roughness sub-layer*”. Fig. 10 (d) illustrates the schematic of velocity profile over roughness with the indication of roughness sublayer, logarithmic layer and outer layer. The roughness sublayer exists beneath the inertial or logarithmic layer and is the roughness counteract of the viscous sublayer. The roughness sublayer is extended to $\approx 2k - 5k$, where k is the roughness height relative to the wall [21]. The utilisation of k_s to define the extension of roughness sublayer instead of k is proposed by Schultz and Flack [237], since it provides a universal measure of the influence of the surface roughness on the mean flow. They showed that the roughness sublayer is extended to $\approx 3k_s - 5k_s$.

4.3. “k-type” and “d-type” roughness

The terms *k*-type and *d*-type roughness are characteristics used for two-dimensional roughness such as spanwise ribs and grooves. The distinction between *k*-type and *d*-type roughness was initially suggested by Perry et al. [295], who observed that for some type of surface roughness the flow behaviour is similar to the usual Clauser or Nikuradse scheme; that is the rough-wall friction factor (f) - Reynolds number (Re) curve for constant U_e is identical to that of smooth walls for constant U (the same concept as that in the wall similarity hypothesis). They also stated that the resulting graph of U^* versus

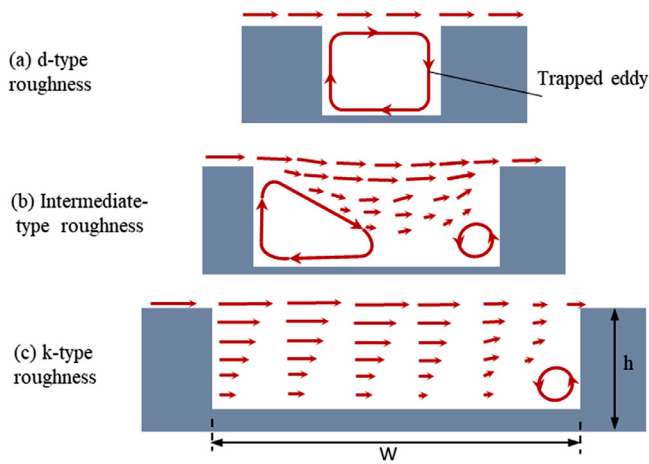


Fig. 11. Schematics of (a) d-type with trapped eddy, (b) intermediate-type roughness with two eddies of different strength as well as separation and reattachment of flow between roughness elements, and (c) k-type roughness with boundary layer velocity structure and a small eddy on the windward side of a roughness element.

kU^*/ν is universal for a given roughness geometry and gives the relationship for both transitionally and fully rough flow regimes in terms of variables required for boundary layer analysis. However, Streeter & Chu [296] and Ambrose [297] have reported examples of different behaviour of flow over rough surfaces, whose friction factor-Reynolds number characteristics are sensitive to relative scale k/d . This type of roughness is characterised by a smooth surface with a series of depression or grooves which the outer flow generate stable vortices. In this case, as Perry et al. [295] showed, U^* is a function of the product $Re\sqrt{f}$ alone which is proportional to dU^*/ν instead of kU^*/ν . Since the characteristic length scale d was used for describing the roughness function, this result appears to be inconsistent with “law of the wall” for rough-wall flows developed by Nikuradse [287] and Clauser [298]. Based on these observations, Perry et al. [295] proposed two types of roughness, k -type and d -type, while the symbols denoting the significant length scale involved. They summarised previous evidence for d -type roughness behaviour and described the related mean velocity profile over d -type roughness. Basically, the d -type roughness is terminology for tightly packed ribs/grooves, while widely spaced ribs/grooves are of k -type roughness. The k -type two-dimensional roughness behaves similarly to the random three-dimensional roughness.

In a follow on from earlier works in this field, Coleman et al. [299] performed experiments of flow structures within the cavities between consecutive ribs of uniform cross-section. They outlined three distinctive types of roughness (Fig. 11) based on the pitch ratio L/H : the d -type, the intermediate type, and the k -type. Here L is the distance between ribs and H is the height of ribs.

Coleman et al. [299] observed that, for closely spaced d -type roughness, stable separated eddies occupy the entire cavity between ribs, with marginal disturbance of the outer flow. However, for widely spaced k -type roughness, reattachment occurs before the next rib, and eddies penetrating to the boundary-layer edge are shed from the roughness. A smooth transition from d -type flow to interactive k -type flow occurs at a rib spacing of $W/h = 5$. The structure of flow over ribs consist of two counterclockwise vortices and the mean flow profile is quasi-logarithmic above roughness tops, while the profile in the interfacial layer below roughness tops changing from exponential (for ribs of $W/h < 10$) to linear ($W/h \geq 10$) to logarithmic ($W/h \gg 10$) with increasing rib spacing.

More recently, Li and Li [300] performed experiments on turbulent open-channel flows over roughness elements. Below the crest, they observed weak eddies in the cavity between adjacent roughness elements for d -type (as illustrated in Fig. 11 (a)), while they indicated two eddies with different strength for the intermediate-type

roughness (as illustrated in Fig. 11 (b)). They had evidence of boundary layer velocity structure for the k -type roughness with a small eddy on the windward side of a roughness element (as illustrated in Fig. 11 (c)). Some researchers have argued that the difference between k -type and d -type roughness relies on the state of the vortex shedding [301]. For k -type roughness, Perry et al. [295] suggested that eddies with a length scale of order k (roughness height) are injected into the flow above the crest of the roughness elements (vortex shedding) which diffuse to the flow further away from the roughness crests, while d -type roughness maintains stable eddies between roughness elements. However, Djenidi et al. [302] and Leonardi et al. [303] have reported vortex shedding into the flow for d -type roughness, although less intense compared to that of k -type roughness.

Jiménez [81] explained the behaviour of d -type roughness as being able to sustain stable recirculation vortexes that isolate the outer flow from the roughness, as illustrated in Fig. 11 (a). The d -type roughness can reduce the drag by trapping pockets of fluid between roughness elements [304] and is especially crucial to heat transfer applications [86]. Ghaddar et al. [305] numerically studied laminar flow over a grooved channel and found that the vortices bifurcate spontaneously to an oscillatory state at fairly low Reynolds numbers, and that the bifurcation eventually leads to subharmonic behaviour in which several grooves eject collectively. Ghaddar et al. [306] later studied DNS of enhanced heat transfer in channels by introducing pulsating flow at frequencies resonating with the natural instability, supporting the idea that similar resonances could occur naturally over d -type roughness. Walls with grooves wider than $\approx 3 - 4 k$ behave like k -type roughness (Fig. 11 (c)) sustaining recirculation vortexes ahead of ribs in the grooves [295,302,307].

Jiménez [81] employed the term roughness solidity (λ) to distinguish between k -type, and d -type roughness. The solidity is a parameter that quantifies the roughness density and is defined as the total projected frontal roughness area per unit wall-parallel projected area. Jiménez [81] outlined two flow regimes: the sparse regime and dense regime. In the sparse regime where the solidity is below $\lambda \approx 0.15$, the impact of the roughness increases with solidity; however, in the dense one, it reduces since the roughness elements shelter each other. The sparse regime ($\lambda < 0.15$) and dense regime ($\lambda > 0.15$) have been reported by other researchers [250,268]. Jiménez [81] reported that flows over the k -type roughness could be characterised as sparse regimes while d -type roughness is in the dense regime category.

The Effective Slope, ES , is connected to the solidity parameter, λ , through the relation $ES = 2\lambda$ [61,72,268]. MacDonald et al. [268] showed that the value of $ES \approx 0.35$, defined as a demarcation point between waviness regime ($ES < 0.35$) and roughness regime ($ES > 0.35$), is associated with a solidity value of $\lambda = 0.175$, quite close to the value $\lambda = 0.15$ (defined by Jiménez [81]). Moreover, Mejia-Alvarez and Christensen [274]; De Marchis [308] have reported that $ES = 0.35$ is a limit between slope dependent and height dependent flows. They also reported that waviness regime can be attributed to the region where $ES < 0.35$ and viscous drag is dominant, while the transitionally rough region falls in the range $0.15 < ES < 0.35$.

In the sparse regimes (k -type roughness), it is readily apparent that the extra drag imposed by roughness should be proportional to the frontal surface of the roughness elements and $k_s/k \sim \lambda$. Despite the utilisation of solidity for distinguishing between different roughness types, it cannot solely fully characterise a rough surface. For example, Jiménez [81] showed a dependency on λ^{-2} , while powers down to λ^{-5} have been also proposed by Dvorak [309]. Other geometrical parameters are required to describe the mutual sheltering of the roughness elements. For d -type roughness, Perry et al. [295] observed that $k_s \approx 0.02\delta$ which shows that k_s is independent of roughness dimensions. Similar results were reported by Jiménez [81], although he also observed inconsistency with a constant value for k_s/δ . Bandyopadhyay [232] reported that turbulent rough-wall flows with $\delta/k < 40$ behave less like d -type roughness; this is also the criterion

where the wall similarity hypothesis is violated and roughness influences the outer layer.

Leonardi et al. [310] suggested that d -type behaviour is obtained when the origin of y is on the plane of the crests, and the frictional drag is much larger than the pressure drag. They asserted that the relative magnitude of the frictional and pressure drag could be used for the distinction between k -type and d -type roughness. However, a fully accurate distinction between roughness types has not been proposed so far. Most of the CFD models that account for surface roughness have been developed based on the observations of the turbulent flow structures over k -type roughness. There is conflicting evidence between turbulent flow over d -type and intermediate-type roughness and the available data from various studies does not point to an agreed conclusion [81]. Therefore, the d -type and intermediate-type roughness should be resolved by the computational grid to obtain accurate results compatible with the physics of fluid over this type of roughness.

5. Methods and correlations for the evaluation of k_s

The equivalent sand-grain roughness k_s , can be assigned to every type of roughness. For this reason, experiments must be performed to find the velocity distribution $U(y)$ in the boundary layer close to the rough wall. Then, it can be written [311]

$$k_s = \exp \left\{ \kappa \lim_{y \rightarrow 0} \left[8.0 + \frac{1}{\kappa} \ln y - \frac{U(y)}{\sqrt{\tau_w/\rho}} \right] \right\}; \quad (21)$$

where τ_w is the wall shear stress at $y = 0$. The flow condition must be checked to make sure that it is a fully rough regime. The value of k_s proposed for a range of regular and technical rough surfaces can be found in Ref. [3,312–314]

5.1. Experimental studies

Most of the correlations of equivalent sand-grain roughness k_s reported in the literature were based on the experimental data of

turbulent flow in fully rough regimes. Several studies have suggested k_s based on the R_a parameter. Among all, Foster [315] showed that ratios of $k_s/R_a = 2$ for typical machined-ground surfaces and $k_s/R_a = 7$ for emery papers with a range of grit size. Koch and Smith [316] found a value of $k_s/R_a = 6$ for emery paper while Bammert and Sandstede [317] estimated a value of $k_s/R_a = 2$ for typical milled surfaces.

However, Bogard [318] showed that using a single roughness parameter such as R_a is insufficient for characterising roughness effects. In a similar result, Goodhand et al. [319] reported that criteria defined only by the R_a parameter are oversimplified. Therefore, several researchers have defined the k_s based on the other roughness features such as; moment of surface height probability density function (PDF) [82,277,320,321], roughness slope [283,322], and density of the roughness elements [280,281].

Musker [321] suggested the use of moments of PDF in the determination of the roughness scale. He then defined the sand-grain roughness height k , in terms of the R_q parameter but modified by the average roughness slope s_p , skewness s_k , and kurtosis k_u as

$$k = R_q (1 + as_p)(1 + bs_k k_u); \quad (22)$$

where a and b are empirical constants to fit the data of naturally rough surfaces. Musker's correlation was developed based on five statistical parameters of rough surfaces and other attempts have been made by researchers to reduce the number of parameters. Townsin and Dey [323] applied another approach to define the sand-grain roughness height k , using the amplitude energy spectral density and a variable long wavelength cut-off to determine the roughness height based on the spectral area of painted ship roughness. Based on the results of 30 experimental sets of ship paint roughness, the best correlation for the roughness height was expressed as [323]

$$k \sim \sqrt{\chi m_0 m_2} = \theta \sqrt{\frac{m_4}{m_2}}; \quad (23)$$

where θ is the spectral area, m_2 and m_4 are the slopes by the second and fourth moments respectively, while $\chi = \theta m_4 / m_2^2$ is the bandwidth parameter. Although this method proved promising in the fully

Table 3
Correlations proposed for calculation of equivalent sand-grain roughness.

Researcher	Roughness type	k_s
Botros [324]	Nikuradse/Colebrook roughness	$(k_s)_{\text{Colebrook}} = 1.306 R_q + 0.078 (R_q)^2$ $(k_s)_{\text{Nikuradse}} = 2.294 R_q$
Dirling [281]	2D roughness including square rods, spheres, cones, hemispheres, right and angles as well as sand-grain roughness	$\frac{k_s}{k} = \{ 0.0164 \Lambda^{3.78}, \Lambda \leq 4.93 \quad 139 \Lambda^{-1.90}, \Lambda > 4.93$
Sigal and Danberg [280]	A dense arrangement 2D transverse roughness including bars, rods, and ribs	$\frac{k_s}{k} = \{ 0.00321 \Lambda_s^{4.925}, 1.40 \leq \Lambda_s < 4.89 \quad 8, 4.89 \leq \Lambda_s \leq 13.25 \quad 151.711 \Lambda_s^{-1.1379}, 13.25 \leq \Lambda_s \leq 100$
Bons [322]	Real turbine blade roughness	$\log(k_s/k) = -0.43 \log \Lambda_{s, \text{modified}} + 0.82$
van Rij et al. [325]	3D irregular roughness including staggered patterns of spheres, spherical segments, and cones	$\frac{k_s}{k} = \{ 1.538 \times 10^{-5} \Lambda_s^{5.683}, \Lambda_s \leq 7.842 \quad 1.802 \Lambda_s^{0.03038}, 7.842 \leq \Lambda_s \leq 28.12255.5 \Lambda_s^{-1.454}, \Lambda_s \geq 28.12$
Bons [322]	Real turbine blade roughness	$k_s = 0.0138 \alpha_{rms}^2 - 0.0261 \alpha_{rms}$
Bons [283]	Real turbine blade roughness	$k_s/R_{ref} = 0.019 \bar{\alpha}_f^2 + 0.0736 \bar{\alpha}_f, \bar{\alpha}_f < 11^\circ$ R_{ref} is the average of several maximum heights measured over subsets of the surface
Boyle and Stripf [320]	Irregular random roughness	$k_s = 4.3 R_q (1 + C_{sk} s_k)$ C_{sk} is constant slightly more than one, and for simplicity, this constant is set to 1.0
Flack and Schultz [82]	Sandpaper [247], honed pipe [326], commercial pipe [291], gravel (medium and coarse) [231], different types of pyramids [249] packed spheres and packed spheres with grid [237]	$k_s = 4.43 R_q (1 + s_k)^{1.37}$
Flack et al. [277]	Rough surfaces with negative skewness	$k_s = 2.91 R_q (2 + s_k)^{-0.284}$
Stimpson et al. [59]	Random roughness fabricated by additive manufacturing	$k_s/D_h = 18(R_a/D_h) - 0.05 \frac{R_{sk}}{D_h} > 0.028$

rough regime, the correlation demonstrated poor agreement with the experimental data on the transitionally rough regime.

Determination of the roughness density requires a full 3D surface map which is hard to achieve for many practical surfaces such as turbine blades, while the surface roughness slope angle α is easier to obtain using 2D trace [283,322]. Bons [322] proposed a correlation for equivalent sand-grain roughness height based on the root-mean-square of local slope α_{rms} . Subsequently, Bons [283] suggest using on the mean forward-facing roughness slope angle $\bar{\alpha}_f$ since it is easier to calculate in practice.

Based on the different methods of roughness characterisation, several correlations for the estimation of the equivalent sand-grain roughness have been proposed in the literature, and are listed in Table 3. The diverse variety of proposed equations in Table 3 demonstrates that a single correlation is not able to estimate the equivalent roughness height or all types of roughness. For example, Flack et al. [278] reported that correlations should be collated based on the positive, negative and zero skewness while Botros [324] suggested that k_s can be calculated by fitting on Nikuradse or Colebrook roughness data using only R_q parameter. On the other hand, Dirling [281] and Sigal and Danberg [280] proposed a combined solidity and shape parameter, Λ , to predict k_s while Bons [283,322] suggested the root-mean-square of local slope α_{rms} .

Flack et al. [278] demonstrated that correlations for waviness ($ES < 0.35$) and highly skewed surfaces must include more surface parameters including effective slope (for wavy surface), shape parameter, and length of the surface elevation. Recent studies have reported promising predictions of rough wall flows using the effective slope, ES , of roughness [61,64,72,249,308,327,328].

Irregular surface roughness is a natural consequence of Additive Manufacturing (AM) processes of metals which have recently received considerable attention among researchers. An example of surface roughness fabricated by AM is illustrated in Fig. 2 (g) in the introduction section. Stimpson et al. [59] studied the surface roughness of channels fabricated by AM. Their attempt to correlate the equivalent sand-grain roughness based on the approaches that existed in the literature (listed in Table 3) was unsuccessful. This was because of the significant difference between the roughness parameters of different test coupons and that the previous studies referred to were performed for external flows where walls did not constrain the boundary layer growth. Stimpson et al. [59] found that one of the simplest statistical measures of channel roughness (R_a/D_h) correlated best with k_s/D_h and also reported that the modified roughness parameter suggested by van Rij et al. [325] is the most appropriate method for rough surfaces fabricated by the laser-based additive manufacturing method [325].

5.2. CFD studies

CFD has been used by several researchers for the evaluation of the proposed correlations in response to irregular and random roughness. The assessment of various correlations for irregular roughness is limited within the literature. Yuan and Piomelli [64] employed the Large Eddy Simulation (LES) method to evaluate the influence of different roughness correlation on the skin friction factor over four distinctive realistic roughness surfaces. They reported that correlations based on the roughness density [280,281] and roughness slope [283,322] can similarly predict the equivalent sand-grain roughness with better accuracy than moment-based correlations [82,277,320,321]. This is mainly because the spectral moments of the surface do not directly include the information on the slope [66]. This statement along with the aforementioned argument of Flack and Schultz [82] concerning the need for information on the roughness density, demonstrate that at least one parameter containing data on the effect of the roughness density or slope should be included in the methods for estimation of the flow characteristics over rough surfaces.

The equivalent sand-grain roughness height is a roughness scale in the fully rough regime, and is not suitable for the transitionally rough flow regime. The DNS method provides the ability to simulate realistic random roughness with a wide range of surface roughness parameters. Thakkar et al. [72] performed a DNS study for 17 irregular rough surfaces at the same friction Reynolds number (the Reynolds number obtained by applying the friction velocity). All surfaces are scaled to the same physical roughness height under the transitionally rough regime. They showed that the solidity (frontal area ratio), skewness, streamwise correlation length, and root mean square roughness height have a significant influence on the roughness function.

According to the above literature review, the equivalent sand-grain roughness appears to be correlated with distinctive sets of geometrical roughness parameters for different rough surfaces, and no universal correlation exists that can predict the flow over surfaces with irregular random roughness. Forooghi et al. [66] performed a DNS study of low Reynold flow over realistic regular roughness. They studied 38 distinctive roughness geometries considering a range of effective slope ($0.2 < ES < 0.9$), skewness ($-0.33 < s_k < 0.66$), kurtosis ($1.9 < k_u < 2.7$), and a new parameter Δ , accounting for the distribution of the roughness element size. They proposed a roughness correlation as

$$\frac{k_s}{R_z} = \begin{cases} \tilde{F}(s_k)\tilde{G}(ES) & \Delta \leq 0.15 \\ F(s_k)G(ES) & \Delta > 0.15 \end{cases} \quad (24a)$$

where R_t is the maximum peak to valley height roughness height over several subsets of the surface. They defined different auxiliary functions $\tilde{F}(s_k)$, $\tilde{G}(ES)$, $G(ES)$, and $F(s_k)$ to isolate the effect of each roughness parameters on the flow. These functions are expressed as

$$F(s_k) = 0.67s_k^2 + 0.93s_k + 1.3 \quad (24b)$$

$$\tilde{F}(s_k) = \begin{cases} F(s_k) & \Delta \geq \Delta_0 \\ F(s_k)[1 + M(\Delta - \Delta_0)] & \Delta < \Delta_0 \end{cases} \quad (24c)$$

$$G(ES) = 1.07[1 - \exp(-3.5 ES)] \quad (24d)$$

$$\tilde{G}(ES) = 1.05[1 - \exp(-3.8 ES)] \quad (24e)$$

where $\Delta_0 = 0.35$ and $M = 1.47s_k^2 - 1.35s_k - 0.66$.

More recently, Jouybari et al. [329] applied Deep Neural Network (DNN) and Gaussian Process Regression (GPR) machine learning methods to develop a high-fidelity approach for prediction of the Nikuradse equivalent sand-grain roughness k_s [329] for turbulent flows over a wide range or rough surfaces. They performed DNS study of 45 surfaces with different roughness topography which differ widely in moments of roughness height (R_a , s_k , and k_u), effective slope, average inclination, porosity, and degree of randomness. Among all surfaces employed in the DNS, 30 of them are identified as fully rough flow regimes. The combination of these 30 DNS results in combination with 15 fully rough experimental data sets of Flack et al. [277]; Barros et al. [330]; Flack et al. [330]; Flack et al. [278] are used for the calculation of k_s using DNN and GPR. Using the correlation of k_s proposed by Flack et al. [277] (see Table 3) as a baseline, the DNN and GPR methods predicted k_s with an rms error of less than 10% and a maximum error of less than 30% which appears more accurate than the correlations existing in the literature. However, they did not suggest any correlation for calculation of k_s .

6. Mean flow over rough walls

As mentioned earlier, current engineering models of the rough-wall turbulent models have been developed based on the extension to the smooth-wall turbulent flow models. This section describes the methods for the extension of the mean velocity and temperature profiles over smooth surfaces to that of rough surfaces. The major impact

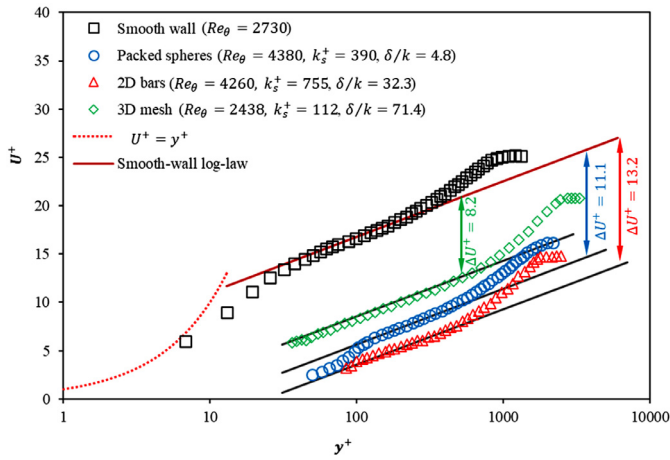


Fig. 12. Mean velocity profiles of experimental turbulent flows over 2D bars and 3D mesh by Volino et al. [223], closely packed spheres by Kim et al. [87], and smooth surface by Wu and Christensen [224]. Downward shift (ΔU^+) of the mean velocity profile below the smooth wall (velocity deficit) is illustrated for each rough surface.

of the roughness is the perturbation of the turbulent boundary layer in such a way that leads to an increase in the wall shear stress, and ultimately, the augmentation of wall heat and mass transfer. This leads to the change of the velocity and temperature profiles near the wall. As shown in Fig. 10, depending on the rough flow regime, the viscous sublayer could partially (in transitionally rough regimes) or completely (in fully rough regimes) be destroyed by roughness. However, according to wall similarity hypothesis, the effect of the roughness on the flow is limited to the roughness sublayer, and the outer layer remains unaffected. In the roughness sublayer, the flow is highly disturbed due to the large turbulence mixing caused by roughness elements, so that three to five roughness heights above the surface is needed where the flow is outside the influence of the roughness elements [21,237,331]. Based on these observations, researchers have attempted to develop models for mean flow over surface roughness, based on the available models of smooth wall flows.

Fig. 12 compares the inner-layer mean velocity profile of smooth walls with different rough walls for zero pressure gradient turbulent boundary layer flow. For the rough surfaces, the similarity in the mean velocity profile is shown for all surfaces, while downward displacement (ΔU^+) of the mean velocity profile below the smooth wall (velocity deficit) is observed. The similarity in the mean velocity profile in the overlap and outer layer indicates that the effect of the surface roughness is limited to the inner layer, which is in accordance with the wall similarity hypothesis.

6.1. Mean velocity profile over rough walls

Clauser [292,298] studied the effect of the surface roughness on the mean velocity distribution and presented a method that has demonstrated to be robust and widely applied. He argued that the velocity profile in the viscous sublayer for rough surfaces follows a logarithmic pattern with a similar slope as for smooth surfaces, as illustrated in Fig. 12. He also found that the sole impact of the roughness is a downward shift of the log-law profile as a function of the roughness Reynolds number, as illustrated in Fig. 12. Clauser [298] and Hama [332] each in an independent study introduced a parameter called roughness function (ΔU^+) to emulate the effect of the velocity shift caused by surface roughness. They observed that the surface roughness shifts the logarithmic profile downwards, commensurate with ΔU^+ , indicating a shortage in the momentum compared to smooth walls. The velocity shift is proportional to $\ln k_s^+$ [333] and is independent of wall distance [334]. The shift velocity and roughness

function is also a result of wall similarity for characterizing the roughness effect on the mean flow. These have been experimentally validated with numerous experiments including the works of Karen A. Flack and Michael P. Schultz and their colleagues [82,223,235–237,241–251] and also Kenneth T. Christensen and his colleagues [87,224,225,252–257] as well as recent DNS studies [71–73,258–260].

The equilibrium boundary layer flow over rough walls comprises two overlapping regions; an outer layer scaling with u^* and δ , and an inner layer scaling with v/u^* and all length scales required to thoroughly characterise roughness (e.g. roughness height, kurtosis, skewness, etc.). It is evident that the velocity profile also depends on the distance from the wall y itself. However, the definition of the origin of y for rough walls requires careful attention, since the roughness itself dislocates the entire flow in y -direction normal to the wall. To account for this, the displaced height is defined as $Y = y - d$, where d is the “fluid dynamic height origin” or “zero-plane displacement” depending on both flow and roughness characteristics.

Clauser [298] proposed the logarithmic velocity distribution for turbulent flow over rough walls as

$$U^+ = \frac{1}{\kappa} \ln y^+ + C - \Delta U^+; \quad (25)$$

where C is the smooth-wall intercept and κ is the von Kármán constant. He also asserted that the roughness function ΔU^+ is a sole function of roughness Reynolds number k_s^+ .

Hama [332] performed an extensive experimental study and proposed the roughness function for fully rough flows as

$$\Delta U^+ = \frac{1}{\kappa} \ln k_s^+ + C - B; \quad (26)$$

Schlichting [76] proved that B is a constant value of 8.5 for fully rough flow regime in pipes. A similar value was reported by Pimenta et al. [335] for fully rough boundary layer over uniform sphere roughness. The relationship proposed for velocity profile by Clauser [298] and the roughness function by Hama [332] are both universal for given roughness geometry in pipes, channels, and zero pressure gradient boundary layer flows. Perry & Joubert [336] demonstrated that this universality extends to boundary layer flows in adverse pressure gradients.

The relationship proposed by Clauser [298] and Hama [332] are confined to the inner layer. By using Coles’ “law-of-the-wake” [90], the correlation can be extended to cover both the overlap and outer region of the boundary layer as [90]

$$U^+ = \frac{1}{\kappa} \ln y^+ + C - \Delta U^+ + \frac{\Pi(x)}{\kappa} \left(\frac{y}{\delta} \right); \quad (27)$$

Flack and Schultz [250] in a review of their 15 years of experimental research of turbulent flow on rough surfaces stated that since no “universal” roughness function exists, no single roughness length scale can characterize all roughness types in all roughness regimes. They also showed that in the fully rough regime, the hydraulic roughness length scale is related to the root-mean-square roughness height and skewness; however, the onset of roughness effects could be associated with the largest surface features which are characterised by the peak-to-valley height. Moreover, the behaviour of roughness function in the transitionally rough regime is complicated and varies significantly between roughness types.

ABL flow over rough terrains such as plant and urban canopies is classified as fully rough since the roughness elements (obstacles) are so large that the laminar sublayer is completely destroyed and the flow is considered to be independent of the molecular viscosity [337]. In ABL flow analysis, the roughness sublayer is generally represented by an aerodynamic roughness length scale y_0 , which matches the smooth wall log-law profile, thus the velocity profile for rough walls reads as [21,338]

$$U^+ = \frac{1}{\kappa} \ln \left(\frac{y-d}{y_0} \right); \quad (28)$$

where d is zero-plane displacement which is required to fix the origin of y . Basically, d is a fluid dynamic property that represents the mean level of momentum absorption by the rough surface [21]. More discussion on calculation of zero-plane displacement d can be found elsewhere [21,233,295,339]. Eq. (28) is only valid in the fully rough regime; therefore, the hydraulic (or aerodynamic) roughness length y_0 is an alternative scale that is similar to the equivalent sand-grain roughness height k_s , imposing collapse to a single profile in the fully rough regime and can be set to $y_0 = k_s/32.6$ as suggested by Schlichting [3].

6.2. Roughness functions

The roughness function (ΔU^+) emulates the effect of the velocity shift caused by surface roughness as a downward shift to the log-law profile. Roughness functions suggested in the literature have been correlated based on the collapse to either traditional Nikuradse [287,288] data with three distinctive functions for each roughness regimes (referred as to “Nikuradse-type” roughness function) or Colebrook et al. [340] data with single expression function (referred as to “Colebrook-type” roughness function).

In the hydraulically smooth regime, the perturbations caused by roughness elements are completely damped out by molecular viscosity, resulting in that $\Delta U^+ = 0$. In the transitionally rough regime, with increasing k_s^+ the roughness elements start to produce drag force, which influences the surface frictional drag and augments the roughness function. In the fully rough regime, the drag on the surface is entirely due to pressure drag on the roughness elements. The limit between these flow regimes depends on the roughness type [341]. The general form of Nikuradse-type roughness function can be written as [342]

$$\Delta U^+ = \begin{cases} 0 & k_s^+ \leq k_{\text{Smooth}}^+ \\ \frac{1}{\kappa} \ln [C_s k_s^+] \times \sin \left(\frac{\pi}{2} \frac{\ln k_s^+ - \ln k_{\text{Smooth}}^+}{\ln k_{\text{Rough}}^+ - \ln k_{\text{Smooth}}^+} \right) & k_{\text{Smooth}}^+ < k_s^+ \leq k_{\text{Rough}}^+ \\ \frac{1}{\kappa} \ln [C_s k_s^+] & k_s^+ > k_{\text{Rough}}^+ \end{cases} \quad (29)$$

where C_s is the roughness constant and depends on the type of roughness. Nikuradse's resistance data for pipes roughened with tightly packed, uniform sand-grain roughness suggests the value of 0.5 for the roughness constant. If the type of roughness departs from the uniform sand-grain, other values should be adapted. Experimental evidence for non-uniform sand-grains, ribs, and wire-mesh roughness, recommends a higher value ($C_s = 0.5 - 10$). Cebeci and Bradshaw [294] recommended: $k_{\text{Smooth}}^+ = 2.25$, $k_{\text{Rough}}^+ = 90$ and $C_s = 0.253$ for traditional Nikuradse roughness function and $C_s = 0.5$ for other type of roughness. The values of $k_{\text{Smooth}}^+ = 3$, $k_{\text{Rough}}^+ = 15$ and $C_s = 0.26$ were proposed by Demirel et al. [343] with fitting the roughness function proposed by Schultz and Flack [248].

White [344] suggested a Colebrook-type roughness function, given by

$$\Delta U^+ = \frac{1}{\kappa} \ln(1 + 0.3k_s^+) \quad (30)$$

Grigson [340] proposed an alternative Colebrook-type roughness function for the entire roughness Reynolds number based on the data of Colebrook and White [289] and Colebrook et al. [340], written as

$$\Delta U^+ = \frac{1}{\kappa} \ln \left\{ 1 + \frac{k_s^+}{\exp[\kappa(8.5 - C)]} \right\} \quad (31)$$

where $\kappa = 0.41$ and $C = 5.25$.

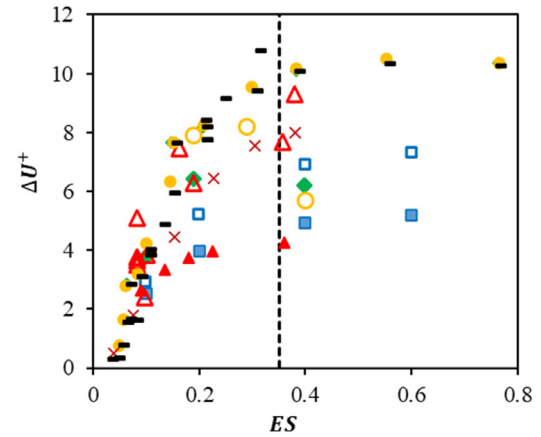


Fig. 13. Variation of the roughness function ΔU^+ with the effective slope ES for different types of roughness. (■) systematically varied rough surfaces with skewness of $S_k = -0.53$ [346]; (□) systematically varied rough surfaces with $S_k = +0.53$ [346]; (▲) closely packed pyramids with $S_k = 0.0$ [249]; (◆) 3D sinusoidal roughness with $S_k = 0.0$ [347]; (Δ) irregular roughness with a range of positive and negative skewness [64]; (●) irregular surface roughness [61]; (×) smoothly corrugated walls with $S_k = 0.0$ [348]; (—) 2D and 3D roughness as well as 2D waves [308]; and (---) $ES = 0.35$.

Reviews on roughness functions can be found in Bradshaw [345] and Flack and Schultz [82] and more recently by Andersson et al. [342].

In dense roughness regime, the roughness function does not scale with k_s^+ ; therefore, an additional factor requires to be taken into account for scaling turbulent flows for all roughness regimes. Fig. 13 illustrates the variation of roughness function ΔU^+ with effective slope (ES) for different type of roughness available in the literature. It is clear that, for sparse roughness regime ($ES > 0.35$), ΔU^+ does not scale with ES ; however, in the dense roughness regime, ΔU^+ decreases with increasing ES , which is consistent with the observation in d-type roughness. The reason for this observation can be attributed to a reduction in the Reynolds shear stress in near wall region that is being pushed away from the rough wall by near wall turbulent cycles [346]. Fig. 13 also shows that, for similar roughness type, ΔU^+ for the surface with positive skewness (peak-dominated surface) are larger than that with a negative skewness (valley dominated surface).

More recently, De Marchis et al. [328] proposed a universal roughness function by performing large eddy simulation (LES) study as well as using other independent data in Refs [61,72,308,347].

$$\Delta U^+ = \frac{1}{\kappa} \ln(ES \cdot k^*) + B^* \quad (32)$$

where k^* is roughness Reynolds number either based on the mean roughness height, k^+ , or based on the root mean square of the roughness k_{rms}^+ . B^* is a constant to fit the data which is equal to 3.5 for the correlation between ΔU^+ and $ES \cdot k_{\text{rms}}^+$, while it becomes 4 for the correlation between ΔU^+ and $ES \cdot k^+$.

6.3. Mean temperature profile

The Reynolds analogy can be used for the estimation of temperature profile over smooth surfaces. However, the Reynold analogy is not maintained for rough surfaces [349–351] since the physics of rough-wall momentum, and heat transfer boundary layers are not similar [352,353].

Several studies have been performed to obtain the mean temperature profile over rough surfaces. Kays and Crawford [78] suggested an alternative simplified formulation for the thermal law of the wall, which has been used for decades. Kays and Crawford's temperature profile reads as

$$T^+ = \Delta T_0^+ + \frac{Pr_t}{\kappa} \ln \left(\frac{32.6y^+}{k_s^+} \right); \quad (33)$$

Where Pr_t is the turbulent Prandtl number. The Reynolds analogy assumption can be replicated by setting $\Delta T_0^+ = 0$, indicating the elimination of the conduction sublayer for the fully rough regime. In fully rough regime, the viscous sublayer in the momentum boundary layer is completely destroyed and vanished while the conduction sublayer still exists in the thermal boundary layer. Therefore, heat transfer over rough surfaces is limited by conduction heat transfer in all rough flow regimes. The ΔT_0^+ is determined empirically and accounts for the conductive heat transfer in semi-stagnant fluid in roughness cavities [78]. Several empirical models for ΔT_0^+ have been proposed in the literature. Dipprey and Sabersky [352] and Owen and Thomson [353] suggested an empirical correlation for the thermal correction ΔT_0^+ as

$$\Delta T_0^+ = \frac{1}{C_{th}} k_s^{+m} Pr_t^n; \quad (34)$$

where C_{th} , m , and n are empirical constants. Dipprey and Sabersky [352] assigned the values of $m = 0.2$ and $n = 0.44$. A commonly used value of C_{th} is 0.8 based on the experiments over closely packed spherical roughness elements. Owen and Thomson [353] determined $m = 0.45$ and $n = 0.8$ for 3D flow across cavities and the value of C_{th} is assigned to 1.92 for randomly shaped roughness. However, Kays and Crawford [78] suggested $C_{th} = 0.8$, $m = 0.2$, and $n = 0.44$ based on the Pimenta et al. [335] experimental study on the closely packed spherical roughness elements. In most ice accretion studies [354–356], the coefficients proposed by Owen and Thomson [353] have been used. Goldberg and Batten [357] suggested $C_{th} = 0.67$, $m = 0.15$, and $n = 0.8$ while they reported that these constants may need small corrections depending on the roughness shape.

The influence of the surface roughness on the pressure drop is more than that heat transfer increase since the heat transfer enhancement is mainly due to the increase in the surfaces heat transfer area caused by roughness [352]. This issue becomes prominent for large roughness elements [358] where Kays and Crawford model fails to completely model the effect of the roughness. Therefore, some researchers have used a surface shape parameter S_{corr} [331,358,359] to account for the increased heat transfer area by rough surfaces compared to that of smooth surfaces. Alternatives to the correlations of Kays and Crawford [78] have been proposed by Ligrani and Moffat [233], Radenac et al. [358], Sucec [360] and Aupoix [359].

7. Roughness effects on fluid flow and heat transfer

As discussed earlier in the paper, roughness can influence heat and momentum transfer in the boundary layer. This influence can be observed through increasing pressure drop, altering flow laminar-turbulent transitions, inducing secondary flow motions, improving flow mixing, and enhancing heat transfer. This section briefly discusses three main roughness effects, comprising roughness induced pressure drop, roughness induced laminar-turbulent transition, and roughness induced secondary flow motions.

Generally, rough surfaces can be classified into two categories [361,362]: 2D roughness and 3D roughness. The term 2D roughness refers to surfaces with a linear profile such as spanwise or streamwise ribs and grooves with different cross-sections which are favourable for heat transfer applications [363–368]. The term 3D roughness refers to surfaces with areal height maps that consist of regular or irregular roughness elements that are systematically/randomly distributed on a surface. A large number of studies have measured the statistical parameters of a wide range of regular and irregular surface roughness fabricated by additive manufacturing [57–59], roughness

generated on the turbine blades [369], and other types of roughness [250,277,341,370–372].

7.1. Roughness induced pressure drop

The form drag and blockage effect induced by roughness elements lead to pressure drop in pipe and channel flows. Early studies focused mainly on the influence of the surface roughness on the pressure drop in pipes and friction losses in boundary layer flows, especially for turbulent flows. Darcy [2] conducted an experimental study of pipe flows with varying roughness, and later, Fanning [373] suggested a correlation for pressure drop in water pipes, including the effect of the wall roughness. These pioneering studies have initiated the field of rough-wall turbulent flows that led to a great deal of theoretical, experimental, and numerical studies in this field.

Nikuradse [287,288] coated the inner wall of pipes with uniformly sized sand grains and investigated the effect of wall roughness on the pressure drop for a range of Reynolds numbers and pipe diameters. Nikuradse found that, although surface roughness has a negligible impact on laminar flows, it considerably influences turbulent flows. He also demonstrated that the rougher the surface, the greater the friction, and therefore a greater pressure loss. On the other hand, for flows with specific combinations of relative roughness and Reynolds number categorised as hydrodynamically smooth regimes, the roughness does not affect the flow friction factor. In hydrodynamically smooth regimes, the average height of the surface roughness elements is much less than the thickness of the viscous sublayer; thus the roughness is not able to influence the turbulent flow over the viscous sublayer. Later, Colebrook [340] conducted experimental studies on the effect of roughness on the flow friction factor in pipes of various diameters, with various linings and roughness ranges. He correlated the results in the well-known Colebrook equation as follows:

$$\frac{1}{f_{Darcy}^{0.5}} = -2.0 \log \left(\frac{k_s/D_t}{3.7} + \frac{2.51}{Re f_{Darcy}^{0.5}} \right) \quad (35)$$

Moody [374] employed the Colebrook equation and plotted the Darcy friction factor " f_{Darcy} " as a function of Reynolds number (Re) for the ratio of equivalent sand-grain roughness height to the tube diameter (k_s/D_t) range of 0 to 0.05, resulting in the well-known Moody diagram that can be used for all flow regimes.

In the transitionally rough regime, the Moody diagram can considerably overestimate the pressure drop as reported by Allen et al. [375] and Langelandsvik et al. [291] for honed steel pipes and commercial steel pipes respectively. This implies that the Colebrook correlation employed in the formulation of the Moody diagram may not apply to a wide range of roughness of engineering interest. Fortunately, from an engineering point of view, this overestimation is on the conservative side. Other shortcomings of the Moody diagram include the lack of an appropriate definition of roughness length scale and the limited applicability of the prescribed surfaces and their roughness properties to many other materials and surfaces. Improved versions of the Moody diagram have been proposed, and LaViolette [376], in a review of the Moody diagram and its alternatives in the literature, proposed a novel alternative to Moody diagram shown to generate a solution without iteration for any type of pipe flow.

The Colebrook equation cannot be rearranged to calculate the friction factor due to the implicit representation of the friction factor " f_{Darcy} " in the formula. Therefore, it can be solved only iteratively or by employing explicit approximation which introduces certain errors associated with calculations. Other empirical relations have been developed that provided good predictions of friction factor without an iterative process. For example, Gregory and McEnery [377] suggested a non-iterative correlation for prediction of friction factor that matches Colebrook's data. Brkic' and Ojbasic [378] applied the genetic

algorithm to optimise explicit approximations of Colebrook's equation to improve the accuracy of prediction. Flack and Schultz [82] proposed a new correlation for the calculation of the overall friction drag coefficient C_f in rough pipes using data on a wide range of surface roughness and roughness elements comprising commercial pipe [291], honed (scratch) [248], sand-grain [247], gravel [231], honed pipe [326], pyramids [249], and packed spheres with grit [237].

The reduction of channel/pipe cross-section area (which is also referred to as channel/pipe size) can increase the effect of the surface roughness on the fluid flow and heat transfer [379–386], specifically in mini-channels (channels with a hydraulic diameter (D_h) between 200 μm to 3 mm) and microchannels (channels with a D_h between 10 μm to 200 μm). Dai et al. [379] reviewed 33 articles, including 5569 experimental data points on liquid flow in rough micro- and mini-channels. They reported no apparent effect on the friction factor and the critical Reynolds number for relative roughness (the ratio of roughness height to the channel hydraulic diameter) lower than 1%. They also found that the geometry of the cross-section (circular, triangular, rectangular) of channels have a marginal impact on the friction factor. Recent studies have shown that roughness can influence laminar flows in pipe and channels [387–389], by increased head loss due to increased dissipation rate [389] and the modification of the development region in internal flows [387]. Ghajar and his colleagues [390–397] showed that increasing the surface roughness contributes to increased friction factor in laminar flow regimes in both micro- and mini-tubes.

Taylor et al. [398] reviewed the historical understanding, current perspective, and possible future development of the effects of surface roughness and texture on fluid flow and reported the limitation of the Moody diagram for relative surface roughness greater than 5%. A new modified diagram was presented for relative roughness up to 14%, which included the constricted flow effects of micro- and mini-channels. Therefore, classical theories of heat transfer and pressure drop for large-size channels has to be revised for channels with hydraulic diameters smaller than 1 mm [382,399–401]. Kandlikar et al. [402] showed that, apart from an increase in shear stress, surface roughness imposes flow area restrictions in microchannels that can further influence the pressure drop. To take these effects into account, they redefined Reynolds number and friction factors based on the “constricted flow diameter”, D_{cf} , as.

$$D_{cf} = D_t - 2k, \quad (36)$$

where D_{cf} is the constricted flow diameter, D_t is the tube diameter, and k is the roughness height. The friction factor based on the constricted diameter is related to the friction factor based on the tube diameter as follow

$$f_{\text{Darcy,cf}} = f_{\text{Darcy}} \left(\frac{D_t - 2k}{D_t} \right). \quad (37)$$

In laminar flows, the Darcy friction factor for circular tubes is expressed based on the Reynolds number calculated using D_{cf} as

$$f_{\text{Darcy,cf}} = \frac{64}{Re_{cf}}. \quad (38)$$

In turbulent flows, the Colebrook equation is redefined based on the constricted flow diameter as

$$\frac{1}{\left(\frac{D_t}{D_t - 2k} \right)^{2.5} f_{\text{Darcy,cf}}^{0.5}} = -2.0 \log \left(\frac{\varepsilon/D_t}{3.7} + \frac{2.51}{Re_{cf} \left(\frac{D_t}{D_t - 2k} \right)^{3.5} f_{\text{Darcy}}^{0.5}} \right) \quad (39)$$

In summary, the Moody diagram and Colebrook correlation overestimate the friction factor [291,375], and for more accurate predictions, relations proposed by Brkić and Ojbašić [378] and Flack and Schultz [82] can be used. The Moody diagram and Colebrook correlation can be applied to predict friction factor in tubes and channels

with small relative roughness, less than 5% [398] and for microchannels, the modified correlation of Kandlikar et al. [402] can be employed.

7.2. Roughness induced laminar-turbulent transition

Many research studies have been devoted to how roughness can influence the laminar-turbulent transition point, which is of significant practical importance. In turbomachines, most of the blades quickly become rough as a result of deposition as well as mechanical or chemical damage. Surface roughness can modify the flow characteristics by introducing flow instabilities which can have a significant influence on the laminar-turbulent transition location [403,404].

In an attached boundary layer under very low-level flow disturbance (free stream turbulence intensity less than 0.5%) natural laminar-turbulent transition occurs [405]. This process is initiated by the generation of exponentially growing 2D waves known as Tollmien-Schlichting (TS) waves (primary instability). This is followed by a 3D instability (secondary instability) leading to significant 3D flows with the formation of streamwise/spanwise vortices. The final stage of such a transition process is the breakdown stage. The large scale vortices break down into smaller structures, generating turbulent spots that eventually merge to form a turbulent boundary layer [76,406].

In the presence of certain sources of sufficiently high flow disturbance, for example, freestream turbulence larger than 1% [407] or surface roughness [408], the laminar-turbulent transition occurs more rapidly, and 2D instability stage of natural transition is bypassed. This type of laminar-turbulent transition is referred to as “bypass transition”. Bypass transition comprises three main regions [407]: the buffeted laminar boundary layer, the region of intermittent turbulent spot formation, and the fully turbulent boundary layer. In the buffeted laminar boundary layer, the shear amplification leads to elongated streaks in the streamwise velocity fluctuations, as shown in Fig. 14. In the intermediate region, the localised perturbations initiate instabilities that eventually evolve into turbulent spots. Once the turbulent spots fully formed, they propagate laterally and grow longitudinally until overtake the main turbulent region. Bypass transition was extensively studied by Paul Durbin and his colleagues [407,409–423]. They investigated mechanisms involved in this type of laminar-turbulent transition and developed related numerical models.

In general, surface roughness moves the onset of laminar-turbulent transition upstream within the flow by bypassing the primary instability. Smaller levels of roughness destabilize the mean flow through a linear amplification of the primary instabilities [425] while larger ones bypass primary instabilities by distorting the flow locally [408,425,426]. Extensive experimental and CFD studies have been performed on the flow laminar-turbulent transition over different roughness, including isolated roughness as well as 2D, and 3D roughness elements distributed on the surface. Durbin [413] reported that, although a significant body of studies has revealed the aspects of laminar-turbulent transition mechanisms due to isolated roughness [276,427–429], bypass transition caused by distributed roughness has not been fully understood. It is found that the low-speed flow behind an isolated element of roughness lifts up from the wall creating an unstable wake or horseshoe vortices. Then, a laminar-turbulent transition is followed by the algebraic amplification of the individual disturbance evolving into streak-like motions. In addition to this, an unstable detached shear layer over the isolated roughness element further accelerate the transition process.

The mechanism of laminar-turbulent transition over distributed roughness is probably related to the formation of streaks by roughness in a similar manner as free stream turbulence flow does. This means that bypass transition may be triggered due to the flow disturbance from the perturbations caused by surface roughness at the wall instead of free-stream turbulence [430]. Fig. 15 illustrates the

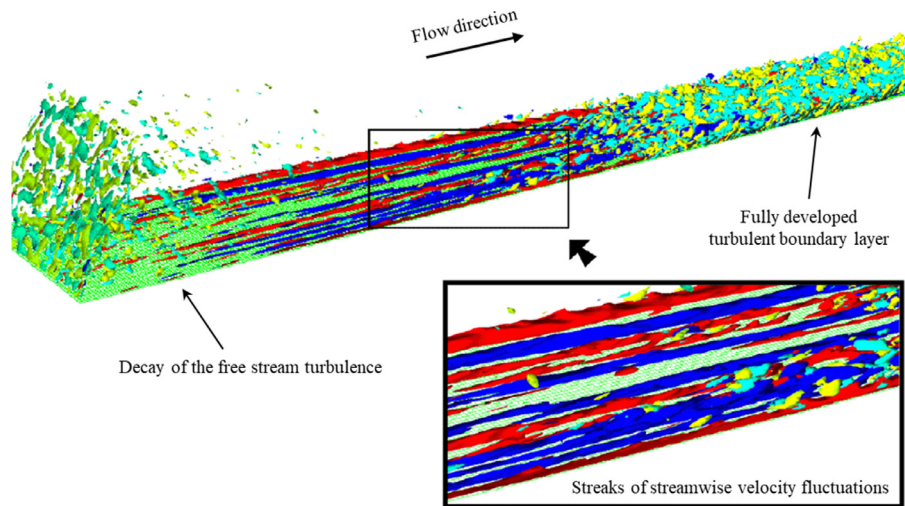


Fig. 14. DNS results of iso-surfaces of the velocity fluctuations due to bypass transition over a smooth surface, adopted from Hughes et al. [424] with the permission of Springer Nature.

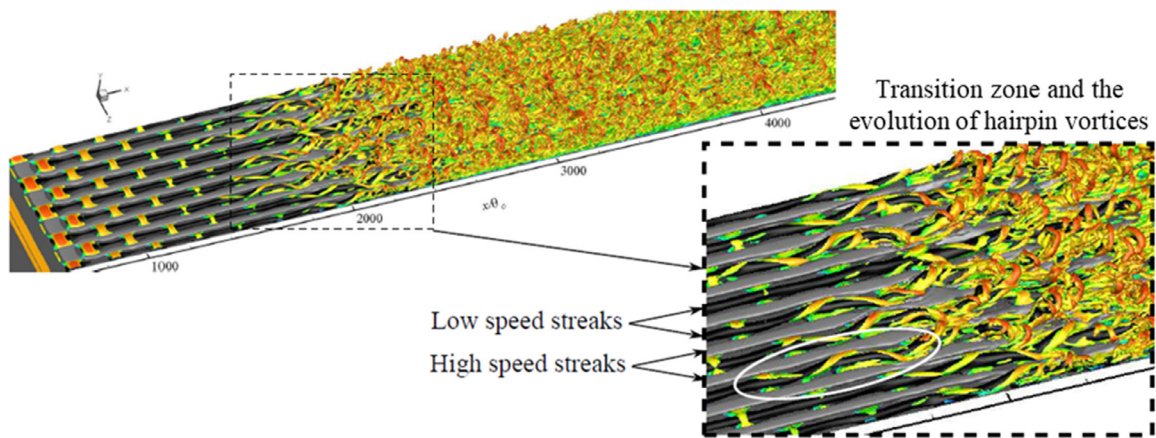


Fig. 15. DNS results of iso-surfaces of the streamwise disturbance velocity for laminar-turbulent boundary layer transition over a surface covered with 3D distributed roughness, adopted from Vadlamani et al. [409] with the permission of Springer Nature.

laminar-turbulent boundary layer transition over a surface covered with 3D distributed roughness, adopted from Vadlamani et al. [409]. It shows that the streamwise streaks are steady earlier and subsequently become unsteady and secondary instabilities develop over the streaks. At this point, hairpin vortices begin to generate, leading to the development of the downstream sustained turbulent flow.

Recent studies have observed the development of a shear layer on the distributed roughness elements. For example, Muppidi and Mahesh [431] reported counter-rotating pairs of streamwise vortices developed underneath the shear layer. The interaction of these counteracting vortices along with the shear layer contribute to laminar-turbulent transition. Vadlamani et al. [409] performed DNS of laminar-turbulent transition over distributed roughness considering the effect of the roughness height. They demonstrated that transient mechanisms alter substantially with the increasing roughness height. For roughness elements that are positioned inside the boundary layer, the elevated shear layer induced by roughness elements changes high- and low-speed streaks near the wall, destabilising the shear layer and promoting a transition to turbulence. For roughness elements larger than the boundary layer thickness, turbulent wakes created behind the roughness elements promote a transition to turbulence. The distributed roughness causes the laminar-turbulent

transition point to advance and increases skin friction or turbulent heating [403]. Due to the transition from laminar to turbulence, the heat transfer coefficient can increase by a factor of 10 [432]. Detailed information on the effect of different distributed roughness on the laminar-turbulent transition can be found elsewhere [319,404,433–435].

Fig. 16 indicates an image of temperature distribution over an aerofoil, recorded by Goodhand et al. [319], which is half rough (sandblasted) and half smooth. The lighter region indicates the turbulent flow, while the darker area identifies the laminar flow. As shown in Fig. 16, it is clear that the roughness causes the location of laminar-turbulent transition to jump to the leading edge in discrete locations referred to as turbulent wedges [319].

Boyle and Stripf [320] showed that, for a fixed roughness height, the transition to turbulence moves forward with increasing Reynolds number demonstrating a reduction in the aerodynamic efficiency of turbine blade cascade. Conversely, at low Reynolds numbers, Boyle and Senyitko [436] reported that, at low Reynolds numbers, surface roughness improves the aerodynamic efficiency of gas turbines by reducing losses at low turbulence intensity; however, at high Reynolds numbers, roughness approximately doubles the vane pressure drop. In a review by Bons [5], a specific region of “roughness benefit”

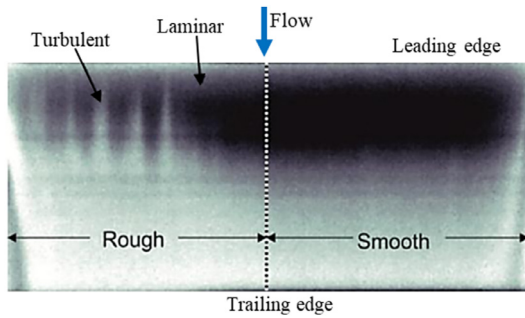


Fig. 16. Premature laminar-turbulent transition from laminar to turbulent caused by surface roughness, adopted from Goodhand et al. [319] with the permission of ASME Publications.

at low Reynolds numbers was reported. This suggests the importance of exploring the influence of the roughness on the fluid flow for enhanced efficiency (both thermal and aerodynamic). More information on roughness induced laminar-turbulent transition can be found elsewhere [429,437].

Surface roughness can also influence the laminar-turbulent transition in microchannels, which is frequently referred to as roughness induced laminar-turbulent transition. Kandlikar and colleagues in various studies [384,385,402,438–449] on microchannels, demonstrated that for high relative roughness, the laminar-turbulent transition occurred at lower Reynolds numbers (as low as 350) and the friction factor was approximately three times larger than predicted by conventional theory. In systematic studies on micro- and minitubes with circular cross-sections, Ghajar and his colleagues [390–397] showed that increasing the surface roughness caused the earlier onset and end of laminar-turbulent transition.

As reported by Schlichting and Gersten ([311] pp. 472 and 473), further effects of the roughness on laminar-turbulent transition could be reflected as flexible wall (inducing elastic wave in the wall), boundary layer oscillations that influence the outer flow, and wave-shaped temperature variations.

7.3. Roughness induced secondary flow

Secondary flows are the occurrence of noticeable fluid motions perpendicular to the main flow direction in wall-bounded flows. Secondary motions of turbulent flow in non-circular cross-section ducts were first observed by Nikuradse [450] with different types of secondary motions then introduced and categorised by Prandtl [451] into two types: first kind, and second kind secondary flows. Prandtl's first kind secondary flow occurs in curved channels/pipes of any cross-section geometry due to centrifugal force which prevails in both turbulent and laminar flows. Prandtl's second kind secondary flow takes place in turbulent flow in non-circular cross-section ducts and is related to inhomogeneities of the Reynolds stresses [452]. It can also occur in plane or symmetrical wall-bounded flows if a local spanwise heterogeneity in wall conditions exists, for example, due to surface roughness.

Despite its weak intensity, secondary motion can significantly influence the flow and alter the mean velocity profile due to the generation of local streamwise vorticity. The spanwise inhomogeneity introduces distinct gradients into the Reynolds stress distribution and alters local flow turbulent properties. These gradients lead to an imbalance of local turbulent production and dissipation rate, which in order to fulfil the balance, inherently translates into the generation of local streamwise vorticity [453,454]. This eventually leads to the formation of large-scale vortex pairs occupying the entire boundary layer thickness, as shown in Fig. 17, and can significantly alter the mean velocity profile. While these large-scale features are strongly evident in the averaged velocity field, the corresponding

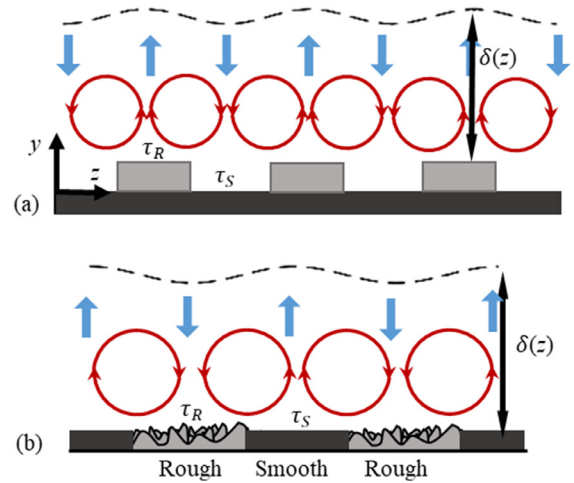


Fig. 17. Schematic of the formation of secondary flows over different types of spanwise heterogeneous roughness: (a) ridge type and (b) strip type. The smooth strips correspond to low wall shear stress (τ_s), while the rough strips correspond to high wall shear stress (τ_R).

instantaneous secondary flow fields do not directly reveal their existence [455]. The application of secondary flow for heat transfer enhancement has been studied by researchers; including improved heat transfer by the generation of the secondary flow of Prandtl's first kind in curved tubes [456–458] and Prandtl's second kind by roughness [368,459–461].

The formation of secondary motions over surfaces roughened with flow-aligned roughness strips in ducts was demonstrated in the pioneering work of Hinze [453]. Later, Hinze [454] hypothesised that the formation of secondary flows is a consequence of the production–dissipation imbalance of turbulent kinetic energy over spanwise heterogeneity. The effect of the secondary flow induced by spanwise inhomogeneity was reported as the thickening and thinning of boundary layer thickness in the cross-plane of the boundary layer [462–466], as shown in Fig. 17. This is due to the formation of low momentum pathways (LMP) associated with upwelling motion and high momentum pathways (HMP) associated with downwelling motions above the surfaces [463,467,468].

In spite of diverse configurations of spanwise inhomogeneity, two main configurations have been well studied in recent decades [69,469,470]: ridge-type roughness and strip-type roughness. Ridge-type roughness consists of spanwise variations of local elevation of the wall, as shown in Fig. 17 (a), while strip-type roughness comprises of spanwise variations in wall-shear stress, as shown in Fig. 17 (b). For ridge-type roughness, the upwelling and downwelling movements were indicated above the elevated and recessed wall areas, respectively [455,464,471–473]. Therefore, secondary flows occur somewhere near the junction between the raised and recessed sections of the surface, with the upwelling either above the junction [470] or above the ridge [466,474,475], as shown in Fig. 17 (a). On the other hand, Strip-type roughness generates upwelling motion over the smooth wall (the low-shear region) and a downwelling motion over the rough walls (the high-shear region) [467,470,476]. Therefore, the secondary flow is placed above the interface between the roughness and smooth surface, as shown in Fig. 17 (b).

Previous studies on ridge-type roughness have investigated flat surfaces with the elevated parts comprising of many possible cross-sectional shapes: rectangular [466,472,477,478], triangular [471,478,479], semicircle [478], trapezoidal [474], streamwise-aligned pyramids [480], and smoother, sinusoidal-like spanwise elevation variations [470,475,481]. Combinations of both ridge-type and strip-type roughness also have been studied in the literature [69,455,464]. Fig. 18 illustrates a schematic of an open channel with

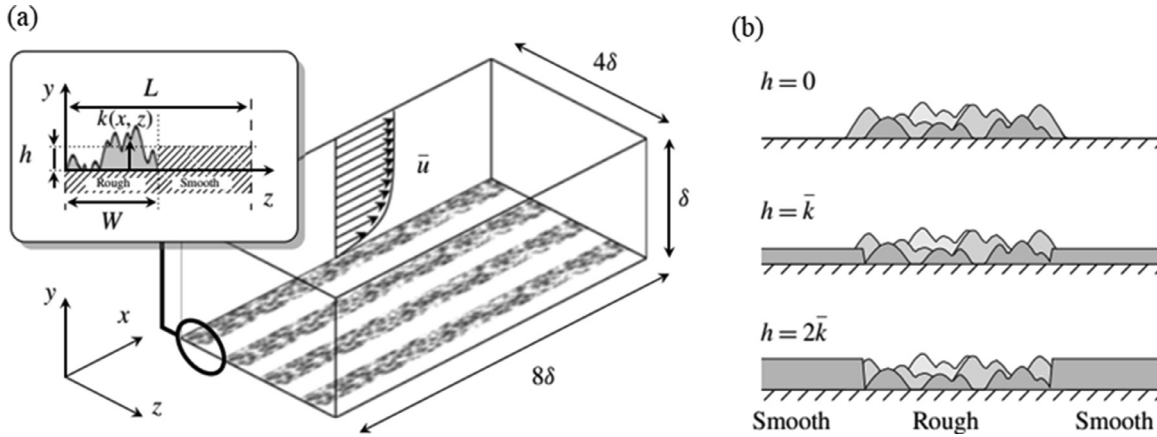


Fig. 18. Schematic of an open channel with combinations of both ridge-type and strip-type roughness, adopted from Stroh et al. [69] with the permission of Cambridge University Press.

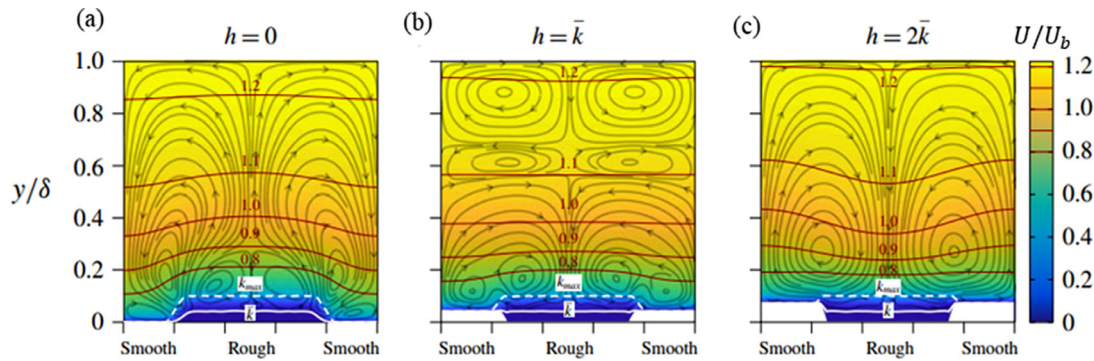


Fig. 19. The profile of mean velocity ratio along the y/δ direction over combinations of both ridge-type and strip-type roughness, adopted from Stroh et al. [69] with the permission of Cambridge University Press. Black lines indicate time-averaged streamlines of secondary motion in the y - z -plane while solid brown lines show the isolines of the streamwise mean velocity distribution. The dimensionless mean velocity changes from zero (blue colour) to 1.2 (yellow colour). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

combinations of both ridge-type and strip-type roughness at the channel's wall used in a DNS study of secondary flow motions by Stroh et al. [69]. Fig. 18 (a) shows the configuration of the open channels, while Fig. 18 (b) indicates three considered elevations of strip-type roughness relative to the smooth surface (channel wall): $h = 0$, $\bar{k}$, and $2\bar{k}$. Stroh et al. [69] observed the formation of counteracting rotations for the same type of the roughness topography, as shown in Fig. 19, depending on whether it is introduced as protruding roughness ($h = 0$) or recessed roughness ($h = 2\bar{k}$). Fig. 19 demonstrates that changing the elevation of the strip-type roughness is an additional parameter for influencing the arrangement of the secondary flow motion from ridge-type ($h = 0$) regime to strip-type regime ($h = 2\bar{k}$). In the intermediate state (Fig. 19 (b)), a more elaborate secondary flow topology is created. In this case, the counteracting vortex pair in the lower channel half are associated with the one observed for $h = 2\bar{k}$ and the small ones positioned in the centre of the roughness with $h = 0$.

Other studies with different spanwise heterogeneity have been studied in the literature; including surfaces with multiple spanwise roughness wavelengths [463], alternating arrangements of directional/anisotropic rough surfaces [465], and converging-diverging riblets [462,465,482]. Converging-diverging riblets also produce secondary flows even in laminar flows [483], where no spanwise gradients in Reynolds shear stress occur and suggest a different generation mechanism to secondary flows of Prandtl's second kind. Experimental observations of the formation of secondary flow over turbine blades with irregular roughness have also been reported (see,

e.g. Mejia-Alvarez & Christensen [274] and Barros & Christensen [463]).

In the numerical simulation, strip roughness can be modelled as spanwise alternating regions of high and low wall shear stress (τ_R and τ_s in Fig. 17) implemented directly as boundary conditions (e.g. Willingham et al. [467]; Chung et al. [476]). A similar method can be used for superhydrophobic surfaces, with spanwise-alternating no-slip and free-slip boundary conditions (e.g. Jelly et al. [484]; Türk et al. [485]; Lee et al. [486]; Stroh et al. [487]).

In practice, the formation of spanwise roughness inhomogeneity exactly aligned with the flow is unlikely. However, this phenomenon can particularly be related to spatially non-uniform roughness formation over turbomachinery blades reported by Bons et al. [369]. The formation of secondary flow motion over damaged turbine blades has been observed by Mejia-Alvarez and Christensen [274] and Barros & Christensen [463]. Strif et al. [459,488] reported induced secondary flow by roughness dominate the heat transfer distribution on the turbine vanes. The spanwise heterogeneous ridge type roughness has been extensively used for drag reduction and heat transfer augmentation, as reviewed by Soleimani and Eckels [30]. They stated that ridge type roughness lifts the large-scale turbulent vortices from the near-wall region and controls the turbulent coherent structures leading to reduced shear stress. The secondary flow motion induced by ridge type roughness enhances flow mixing and increases heat transfer. In river flows, large-scale bed elements elongated parallel with the primary flow is characterized by ridge-type and/or strip-type roughness [470,489,490]. Such bed forms induce secondary

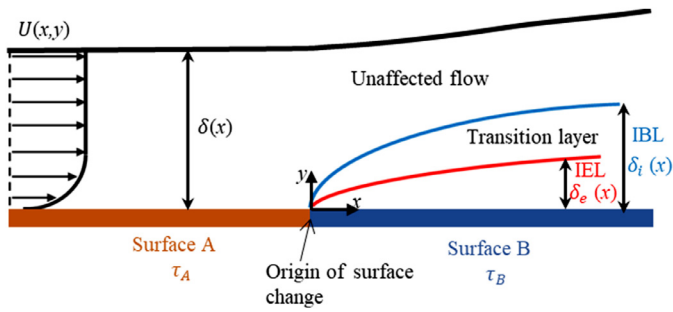


Fig. 20. Schematic of the formation of internal boundary layer (IBL), internal equilibrium layer (IEL), and transition layer for boundary layer over a change from surface A to surface (e.g. RTS or STR). $\delta(x)$, $\delta_e(x)$ and $\delta_i(x)$ represent the boundary layer, IEL, and IBL thicknesses, respectively.

flows and trigger lateral sediment transport, boosting and maintaining longitudinal bedforms [491–493].

7.4. The effect of streamwise roughness inhomogeneity

The inhomogeneity in roughness distribution can occur in spanwise direction (as discussed in the previous section), streamwise direction [494,495], or combination of these surface changes [496–498]. Streamwise roughness heterogeneity prevails in numerous engineering wall-bounded turbulent flows. For example, the roughness condition changes occur at the interface between forest and grassland or the bio-fouled patches on the hull of a ship. Such streamwise heterogeneities can take place in a wide range of configuration, among which rough-to-smooth (RTS) and smooth-to-rough (STR), as shown in Fig. 20, are mostly studied in the literature.

The surface change triggers various phenomena that lead to the deviation of the near-wall flow from the equilibrium state. The near-wall flow accelerates due to the RTS surface changes [495,499], while it decelerates due to STR surface changes [494,500]. The near-wall flow influenced by the new surface is known as Internal Boundary Layer (IBL) [501–503] and is characterised by a thickness of $\delta_i(x)$. The flow away from the wall remains unaffected by surface change and still carries the history from the upstream surface (Fig. 20). The lower part of the IBL that has reached equilibrium with the new surface is referred to as Internal Equilibrium Layer (IEL) and is characterised by a thickness of $\delta_e(x)$. The influence of the new surface condition starts from the beginning of the surface change and gradually propagates towards the interior of the flow with transitioning downstream of the flow.

In the literature, flow recovery after streamwise-varying roughness has been studied theoretically [504–507], numerically, and experimentally. Wind tunnel experiments have been conducted for flow over a fabricated RTS and STR surface change with roughness geometries composed of square bars [494,495,500,508], mesh roughness [509–511], transversely grooved surface [512], or grit roughness [511]. Krampa-Morlu and Balachandrar [513] performed an experimental study of RTS flow recovery with d-type and intermediate-type rib roughness. They showed that the intermediate-type roughness had a higher roughness effect. Cheng and Castro [514] conducted an experimental study of flow field downstream of a step change, where the roughness height becomes ten-fold of the downstream roughness.

In the literature, related computational studies have been performed using roughness modelling such as wall-modelled LES, RANS as well as direct resolving of roughness with DNS. The widely used wall model for LES is the equilibrium logarithmic-law of the wall [163,515], and its extension to non-neutral flow using M-O similarity [516,517]. Saito and Durbin [518] developed a wall-modelled LES to investigate the turbulent flow transition over a smooth-rough-smooth step change. They reported an overshoot and undershoot

immediately after the step change, which has been predicted before by other researchers [504,505,519–521]. A few studies have conducted direct resolving of roughness using DNS of flow with different roughness configurations, including 3D sinusoidal surface [522], rib roughness [523–525], and cube roughness [430,526].

Flow recovery over streamwise-varying roughness involves various flow phenomena, two of which are widely addressed in the literature: (i) the departure from equilibrium, and (ii) the formation of an internal boundary layer (IBL).

7.4.1. Departure from equilibrium

Generally, the majority of the flow within IBL is still non-equilibrium with the local wall condition, and a small portion of δ_i is taken up by δ_e (approximately 5 %) [527,528]. Most of the experimental and numerical studies have evaluated the flow characteristics from a certain y^+ , assumed as departure from equilibrium. Among all, Carper and Porte-Agel [509] conducted an experimental study of RTS surface change at $Re_\tau \approx 8800$, considering the first measurement point at $y^+ = 88$. Antonia and Luxton [495] and Hanson and Ganapathisubramani [511] studied RTS surface change at $Re_\tau \approx 1700$ with performing the first measurement at $y^+ = 95$. Jacobi and McKeon [508] conducted the measurements down to $y^+ = 3$ for an RTS surface change at $Re_\tau \approx 970 - 1200$. They assumed that the viscous sublayer departed from equilibrium. For high Reynolds number flows, the IEL can be extended to the larger y^+ . For example, Saito and Pullin [518] performed an wall-modelled LES study of RTS surface change with $Re_\tau \approx 2 \times 10^4 - 2 \times 10^6$, considering the first grid at $y^+ = 410$. Silva-Lopes et al. [515] conducted wall-modelled LES at $Re_\tau \approx 1.5 \times 10^5$ with the first grid point at $y^+ = 260$. Rouhi et al. [522] conducted DNS studies of RTS and STR surface changes and showed that log-law assumption for the smooth patch is valid after a higher y^+ compared to that of rough patch. They found an error of approximately 5 % compared to experiments for velocity profile if the log-layer assumed positioned at $y^+ = 30$. Ismail et al. [526] performed a DNS study of RTS surface change at $Re_\tau = 920$ with cube roughness and observed the characteristics of smooth wall below $y^+ = 50$.

7.4.2. Internal boundary layer

One of the most important representatives of flow recovery after a surface change is the thickness of the IBL. Various definitions of the IBL have been proposed in the literature (Table 1 of Savelyev and Taylor [503]). In the majority of studies, the growth of δ_i with the streamwise location is described by $\delta_i = \alpha x^b$ with various values of b ranging from 0.2 to 0.8 proposed based on the adopted definition of IBL. Those studies that employed similar definitions have obtained close values of b . For example, Cheng and Castro [514] and Lee [525] employed the definition of Pendergrass and Arya [529] for an STR surface change at $Re_\tau = 2500$ and obtained close values of b (0.33 and 0.22). Antonia & Luxton [494] and Win et al. [530] used the definition of Antonia & Luxton [494] for a STR surface change at $Re_\tau = 2200$ and 2600, and achieved values of 0.72 and 0.8 for b respectively.

Li et al. [531] observed the recovery of the viscous region almost immediately to an equilibrium state with the new wall condition. Rouhi et al. [522] observed that the buffer region and beyond takes 5δ (over the smooth patch) and 2.5δ (over the rough patch) before recovering to equilibrium conditions. Ismail et al. [523] observed a slow recovery of mean flow and Reynolds stress profile. They divided the flow after an RTS surface change into two zones: (i) initial strong non-equilibrium zone ($0 < x/\delta < 0.5$) with sharp change in skin friction proceeding a weak non-equilibrium zone ($x/\delta > 0.5$). Ismail et al. [526] suggested a length of an order 50δ over the smooth surface after a RTS surface change to recover the Reynold stress. Van Buren et al. [532] studied flow recovery of turbulent flows after RTS surface change in pipes. They reported the recovery of pressure gradient, skin friction and mean streamwise momentum after approximately 20 radii downstream of the surface change, while the turbulent

stresses is recovered after about 120 radii downstream of the surface change.

Rouhi et al. [524] conducted DNS of turbulent flow over RTS and STR step changes using a 3D sinusoidal surface (egg-carton), considering the roughness origin was placed aligned with (no-step), below (step-up) and above (step-down) the smooth patch. They studied the effect of the roughness height of $k_s^+ \simeq 106 - 160$, as well as a range of $Re_\tau \simeq 650 - 700$ over the rough patch and $Re_\tau \simeq 400 - 450$ over the smooth patch. They found that changing the roughness origin and Re_τ significantly influence the departure from equilibrium, while the effect of the roughness height was marginal.

Although extensive studies have been conducted in this field, turbulent flow over roughness inhomogeneity is far from being understood. For example, limited type of roughness has been considered in the literature, and recording experiments for a wider range of roughness is required. No study has been performed with multiple patches of roughness, which can have a particular application in naturally formed biofouling on ship hulls. Although wall-modelled LES has shown promise in modelling these type of flows, a wide gap for developing numerical models for such flows exists. Almost all of the previous studies have concentrated on the flow characteristic of such flows without investigation of passive scalar transport that can significantly contribute to heat transfer applications and the study of air pollutant dispersion.

8. Conclusion

Turbulent flow over rough surfaces has been studied for about a century through a great deal of experimental and CFD studies. Roughness can influence fluid flow and heat transfer by increasing pressure drop, altering flow regime laminar-turbulent transition, and by inducing secondary flow motions which lead to an enhancement in flow mixing and heat transfer.

In most cases, roughness can be quantified by a single scaling parameter, known as equivalent sand-grain roughness height k_s , which can be expressed based on the statistical parameters of surface roughness (R_a , S_{10} , k_{10} , ES , etc.). Empirical correlations for the calculation of k_s were reviewed in this paper and it was found that methods using only a single parameter or height (R_a or R_q , etc.) were unsuccessful in modelling flows over rough surfaces. Therefore, a series of studies have included the effect of roughness density in the formulation of k_s , and alternatively incorporating surface moments such as s_k and k_u owing to the difficulties in measuring the roughness density in irregular roughness. However, no universal correlation has been proposed so far. The selection of an appropriate correlation for k_s is difficult due to the variety of the proposed correlations, which can vary up to a factor of 5 [5]. Other reasons are that k_s is a roughness scale in the fully rough regime which is not necessarily useful in the transitionally rough regime, and the wide diversity of roughness types. In addition, the dense roughness regime, which is associated with d-type roughness, does not scale with k_s . Therefore, as stated by Flack [286], it is time to move beyond characterising roughness by k_s . Another roughness scale is required that can describe the onset of roughness effect and transition to the fully rough regime as well as the shape of roughness function in all roughness regimes.

Fluid flow models, particularly those that account for surface roughness, have been developed based on the structure of turbulent flows. Numerous experiments have been conducted, and several theories have been hypothesised to describe flow structure over smooth and rough surfaces. The Monin–Obukhov (M–O) similarity hypothesis was one of the initial theories that were proposed based on the mixing length theory to describe vertical distributions of mean flow velocities and temperature using “universal functions”. Experimental studies suggest that M–O similarity is about 10–20 % accurate and can merely be applied to the active region of turbulent flow [129]. On the other hand, Townsend hypothesis thoroughly explains large-scale

motions near smooth in both inner and outer layers and provides a conceptual model of boundary layer kinetics and eddy motions. The wall similarity hypothesis is an extension of the Townsend’s Reynolds number similarity to rough-wall flows, which has been verified by numerous experimental and DNS studies for various roughness regimes and flow conditions, including separated flows. Wall similarity explains that the effect of the roughness is confined in the roughness sublayer and the outer region remains unaffected by roughness. However, evidence of roughness effects into the outer layer has also been observed in several experiments where the ratio between the thickness of boundary layer and roughness height were smaller than 40 [81,235]. Conversely, some other researchers found similarity at lower ratios.

Despite progress in finding criteria for the transition from k-type to d-type roughness, a fully accurate distinction between roughness types has not been proposed so far. The proposed criteria include state of the vortex shedding, roughness solidity λ , and effective slope ES . The k-type roughness injects a vortex into the flow above the crest of the roughness elements (vortex shedding) which diffuses into the flow further away from the roughness crests. However, weak vortex shedding has also been observed in d-type roughness. Other studies have associated the k-type with sparse roughness regimes ($\lambda < 0.15$) and the d-type with dense roughness regimes ($\lambda > 0.15$). A number of recent studies have observed that at effective slopes $ES \approx 0.35$, that is a delineation between waviness regime ($ES < 0.35$) and roughness regime ($ES > 0.35$), is associated with a solidity value of $\lambda = 0.175$, approximately close to the value $\lambda = 0.15$. The available data from various studies concerning turbulent flow over d-type and intermediate-type roughness does not point to an agreed conclusion or trend, which is mainly related to sheltering effect introduced by these types of roughness.

The distribution of mean velocity over rough surfaces can be characterised as a downward shift in mean velocity from the smooth case using a roughness function (ΔU^+) which is solely a function of roughness scale k_s^+ . However, correlating ΔU^+ with the roughness scale is only valid for sparse roughness regime (k-type roughness). The dense roughness regime (d-type roughness) does not correlate with the roughness scale, whereas it can be scaled with the effective slope (ES). Recent studies suggested employing both roughness scale and effective slope to be able to universally scale turbulent flow in all roughness regimes.

In general, roughness can cause an early transition from laminar to turbulent through bypass transition. Although a significant body of study has cast light on the mechanisms of bypass laminar-turbulent transition due to isolated roughness, bypass transition caused by distributed roughness has not been fully understood. However, no apparent effect on the friction factor and the laminar-turbulent transition properties (critical Reynolds number) for relative roughness lower than 1% has been reported [379].

Numerous studies on turbulent flow over rough surfaces have been devoted to heterogeneous roughness which contributes to important engineering applications including heat transfer, ABL flows, aerodynamics, hydraulics, etc. Most studies have investigated streamwise inhomogeneities which cause a transition in the boundary layer, and spanwise inhomogeneities which contribute to the generation of secondary flow motions, while a few studies investigated a combination of both. Although recent experimental and DNS studies have shed light on the different aspects of flow recovery over spanwise surface changes, this area of research is not fully understood. Spanwise heterogeneity resulting from surface roughness have demonstrated the formation of the secondary flow of Prandtl’s second kind in the turbulent boundary layer. Many of the recent DNS studies have been devoted to investigating this type of secondary flow induced by roughness. The formation of secondary flow is related to the Reynolds stresses’ inhomogeneities and leads to the thickening and thinning of boundary layer thickness in the cross-

plane of the boundary layer. There is a wide research gap for investigating scalar transport over both spanwise and streamwise heterogeneous surfaces.

The main conclusion of this review paper is that despite many advances being made in respect to the fundamental understanding of turbulent flows over rough surfaces, this subject is far from being comprehensively understood. In almost every study or theory, supporting and conflicting evidence can be found. The main reason is related to the chaotic and random nature of turbulent flows that make accurate predictions almost unattainable. Other reasons are a lack of sufficient systematic studies on the structure of turbulent flow and because of the diverse variety of roughness which significantly impacts the dynamics of the flow in roughness sublayer.

Recent advances in additive manufacturing technology, have increased the possibility of fabricating artificial engineered roughness ranging in size from macro to nanoscale [54,112]. This technology provides the designer with the ability to control the shape and pattern of the roughness elements and can be used for the fabrication of roughness to perform systematic experiments. The significant computational power of modern computers and high-performance computing have provided the ability to perform DNS of turbulent flows over realistic roughness. This active area of research seems set for further investigation. Advances in these technologies coupled with a complete synthesised and thorough in-depth review of the domain area, as presented by this paper, can help bridge the knowledge gap in our fundamental understanding of turbulent flows over rough surfaces.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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